

Title: IN - Integrative Neuromedicine

1. General information

1.a Applicants:

Name*	Institution	Department	Email	Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead/ Applicant]
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Corresponding Lead: Daan Brinks
Planned start date of seed funding: 01/09/2022
Primary Convergence theme: Fundamentals of Health and Disease.
Secondary Convergence theme (*if applicable*): Improving Health Journeys
Keywords: neurodevelopment in health and disease, functional assays, mTOR pathway, normality, high resolution imaging, digital twin, ultrasound

1.b Summary (max 300 words):

Over 250 million people suffer from Neurological Disorders (NDs) globally, of which approximately 41 million live in Europe. Treating NDs accounts for 35% of Europe's total disease burden with a yearly cost of €798 billion¹⁻³. Yet comprehensive treatments for NDs are lacking.

NDs are complex at many levels: from molecular mechanisms through cellular manifestations and brain-wide effects, to the experience by individual patients and societal interpretation and acceptance. Current large-scale efforts to understand or treat NDs therefore rarely meet with complete success or universal acclaim. In this flagship, we propose to lay the groundwork for an integrative multidisciplinary approach that is necessary to understand the intricacies of ND manifestation, from the cellular to the phenomenological level. We will investigate NDs across scales, from molecule to society; across time, from conception to death; and across approaches, from biological to philosophical.

This flagship aims to create the first-of-its-kind cohesive and synergistic pipeline for the investigation and treatment of NDs. Through this pipeline we will accelerate the translation from therapeutic ideas to the clinical stage in the next 5 to 10 years. Concretely we will focus on the mTOR pathway, which is central to many NDs, including autism spectrum disorders and epilepsy.

The project consists of three Work Packages: **Mapping**, which focuses on techniques to acquire fundamental information about the brain throughout life; **Understanding**, which seeks to inquire, at every level, the difference between normal and diseased brains; and **Intervention**, which aims to translate findings to effective and accepted diagnostics and treatments.

The project will fundamentally change the way NDs are being researched and treated. We will increase efficacy and usefulness of fundamental insights; increase the speed of bench to bedside translation; and increase the success and acceptance of proposed interventions, with the final aim of significantly decreasing the impacts of NDs.

1.c Public Summary (max 300 words)

Over 250 million people suffer from Neurological Disorders (NDs) globally, of which approximately 41 million live in Europe. Treating NDs accounts for 35% of Europe's total disease burden with a yearly cost of €798 billion. Yet comprehensive treatments for NDs are lacking.

Unraveling the mysterious workings of the brain in health and disease is the holy grail of today's neuroscience and medicine. It will be impossible to propose new medical treatments and interventions to mitigate neurological disease symptoms and impairments without such knowledge. The enterprise to understand the brain calls for an interdisciplinary team able to study

it throughout all levels, from the very small (i.e., molecules and cells) to the very large (i.e., brain networks), while keeping the human subject's unique experiences and needs in perspective.

To achieve this goal, our flagship team includes fundamental scientists, engineers, philosophers, psychologists, and clinicians across the three different institutions to converge towards a personalized integrative medicine approach. We want to combine data from all scales, from birth to deathbed, to understand the evolution of neurological disorders. We will start with identifying common biomarkers involved in autism and epilepsy. Our research will enable early diagnosis and improve potential treatments over time. Furthermore, we will show how new technology can be developed and used to achieve medically effective personalized treatment, while providing a comfortable and satisfactory experience from the individual's perspective.

Our flagship project's effectiveness, cohesiveness, and sustainability lie in the complementary ingenuity of interdisciplinary team members, our extensive network of academic and industrial partners, and the multiple sponsors seeking groundbreaking research. Ultimately, we will establish a pipeline that eases the transition from the bench to bedside through various breakthroughs to improve the subjects' quality of care and quality of life.

2. Relevance to Convergence Health & Technology

mission (50%) – max. 2500 words for 2.a to 2.e [2447 words]

2.a Contribution to H&T mission and fit with H&T theme(s)

The proposed flagship integrates the core concepts of Convergence H&T:

1. Lifelong health, from pre-birth to end of life. Central to our approach is the mapping, understanding and intervention of neurological disorders related to the changing brain. We aim to integrate knowledge from pre-birth to old age and from the molecular to the existential level, throughout the patients' health journeys.

2. Socio-economic equality in health(care). Our integrated approach aims to deliver personalized medicine with equal access, resulting in a healthier lifespan for all. The information this approach provides would need to become an accepted part of patients' health journeys. Integration and acceptance of the new types of data and insights that advanced technology can bring, by clinicians and patients, is a central part of our flagship, spearheaded by our applicants from EUR.

3. Prevention and early diagnostics to proactively maintain health &

4. Individually tailored precise medical treatment for all. Our goal is to develop early diagnostics and improve personalized medicine, while maintaining cost efficiency. For a system as complex as the brain, this will rely heavily on the ability to create comprehensive and reliable models and pipelines for analysis, prediction and intervention, supplemented with detailed diagnostic tools. We aim to create biological models on different scales (cellular to network/tissue from patients and animal models) complemented by comprehensive virtual computer models of the human brain, (i.e. digital twins) that will allow us to understand, diagnose and treat the changing state of the brain and identify deviations from normality. The systemic, integrated approach will be central to allowing these convergence goals to fully flourish.

The proposed flagship objectives fit with the Convergence H&T themes:

Our proposal is well aligned with the *Fundamentals of Health & Disease* theme, as **WP1: Mapping** and **WP2: Understanding** look at the molecular basis of life and disease across different spatial and temporal scales. Concurrently, we will develop new biological model systems, instrumentation, data analysis and simulation methods.

These Work Packages (WPs) will fuel our own "convergence knowledge pipeline" to feed into **WP3: Intervention**, where we aim to improve the translational pipeline for therapeutic interventions. This will effectively connect basic research to new healthcare innovations. We would, therefore, also contribute to *Improving Health Journeys* by offering possibilities for early disease detection and personalized treatment for optimal outcomes.

The proposal brings together a new generation of assistant and early career associate professors at TU Delft, Erasmus MC and EUR. We are building on the *Delft/Rotterdam Neuroscience and Technology workgroup* established in 2018 and will expand it into a virtual institute focused on Integrative Neuromedicine. We are experts from the medical sciences, life sciences, technical and computer sciences, social sciences, and humanities with a common interest in neuroscience and medicine. Convergence of this wide range of disciplines will lead to disruptive innovations leading to healthier lifespans, and pave the way towards groundbreaking results, not feasible at the scale

of one single institution. Our virtual institute will enable training of a new generation of highly interdisciplinary PhD students, postdocs and PIs that require the knowledge of all three institutions to realize their ambitions.

2.b Synergy between disciplines

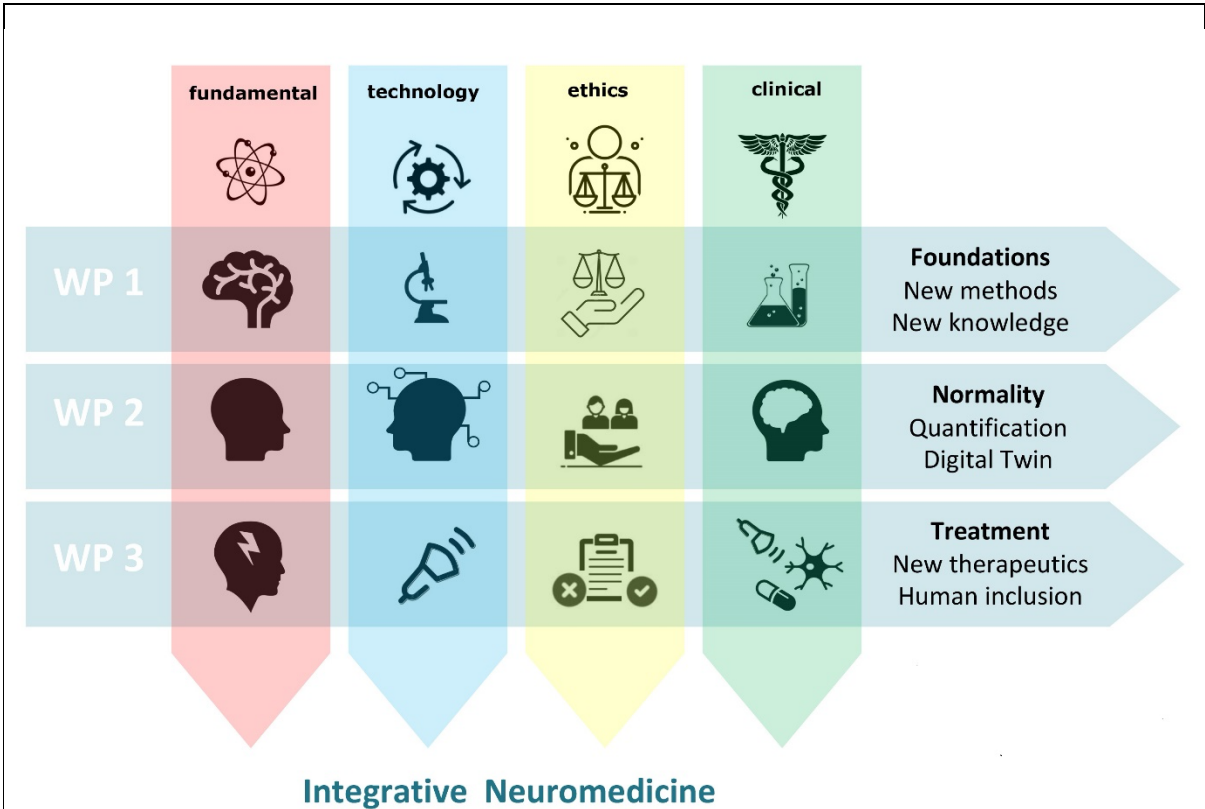


Figure 1: The defined WPs require synergistic approaches between disciplines, and WP results feed into each other, together creating healthier lifespans for all through Integrative Neuromedicine.

We envision a data-driven scientific and engineering approach through the process of data collection in the lab and the digitalization of time-evolving models from cradle to grave. These models will integrate in vitro and in vivo fundamental and pre-clinical research, as well as in silico and clinical data. This unified approach is aimed towards quantifying normality through biological and phenomenological biomarkers.

Our interdisciplinary team of scientists, engineers, philosophers, and clinicians will focus on the characterization of pathologies associated with the mammalian Target of Rapamycin (mTOR) pathway, central to common neurological disorders like autism and epilepsy^{4,5}. Disruptions to this pathway has ramifications from the cellular to the phenomenological manifestation of neurological diseases⁶⁻⁹. Therefore, scientists and engineers will work side-by-side to unveil new knowledge on Biology, Biophysics and Biodynamics (B³) in health and disease. This knowledge is key in producing computational models (digital twins) validated in vitro and in vivo throughout aging, spanning the entire progression of neurological disorders, enabling earlier diagnosis, prognosis and intervention. Philosophers will work with scientists and engineers to guarantee

ethical development of patient treatments and safeguard their needs and priorities. In particular, they will have a key role in the quality-of-care validation of the proposed models from the patients' unique experience and in the fundamental reflection on what counts as normal or pathological.

We have defined WPs that purposefully bridge disciplines (**Figure 1**): in **WP1: Mapping** the overarching goal is to develop technological tools (e.g. microscopy, imaging sensors, organoids, computational models) to study the brain using complementary approaches across life stages from early development to old age and thereby gain new fundamental knowledge. In **WP2: Understanding** we ask the question what a normal brain is. Again, this question will experimentally be approached across scales and life stages, linked together by theoretical and computational descriptions. Concretely, this means that insights from WP1 will be caught in models that will be compared to experimental disease models at the cellular, behavioral and patient level, always with an eye on the normative question of what should and should not be treated. These insights, both at the fundamental and translational level, will feed into **WP3: Intervention**, where we will develop multilevel interdisciplinary treatments. Specifically, we will focus on novel drug delivery systems, non-invasive brain stimulations and psychological approaches.

The pipeline, from mapping the brain, via understanding the normal and abnormal brain, to efficient intervention, including fundamental philosophical reflection and ethical evaluation at all stages, is uniquely challenging. Therefore, it can only work if all disciplines represented in the consortium converge on this common goal. Investigating disease models with this pipeline in a targeted, integrated way, benefiting from the expertise of all the consortium members, will make the convergence partners world leaders in Integrative Neuromedicine.

2.c Long term sustainability plan

Feasibility of the program

The Integrative Neuromedicine team has the required ingenuity and expertise (see **Figure 7 section 4b**) to achieve the main objectives. We are a new generation of internationally recognized PIs, who already have impressive funding resources (see **Figure 6 section 4a**), widespread facilities that are required to achieve the program's objectives (such as multiple spaces for cell culture, animal housing, experimentation, and computation, see 3.b), great embedding in their host departments/institutions, and a multitude of collaborations with academic and industrial partners.

Long term sustainability plan beyond the funding period

Our virtual institute builds on core principles of the *Delft/Rotterdam Neuroscience and Technology workgroup* established in 2018. Since then, we have already demonstrated commitment to community building across all three institutions, with regular joint meetings, shared training opportunities and collaborative projects. With the Convergence Program support, we will expand on this with the creation of shared facilities and collective partnerships with industry.

Our general research topic is well-aligned with funding opportunities of the NWO-NWA routes (e.g., Measuring and detection, NeurolabNL, and Personalized medicine) and European programs (E-Brains; Horizon Europe strategic plan, Cluster 1: Health). In our virtual institute, consortia can go for small to medium scale impact-based collaborative grants (NWO-M/XL) and acquire the

critical mass to apply to grants such as NWO Gravity, ERC Synergy and EIC Pathfinder. To translate our fundamental research into the promised societal impact, we will consider suitable “Kennis en Innovatie convenant” (KIC) options (e.g. with Open Technology program, ZonMW, Hoofdzaken, and connections with patient associations).

In the long term, the treatment development pipeline outlined in this program will lead to interventions with the explicit goal of valorization in collaboration with patients, clinicians and industrial partners. We will realize this either via licensing agreements or via the creation of startups, which will be leveraged by the virtual institute funding.

Finally, our team of early career academics represents a broad spectrum of talented scientists in cutting-edge research areas with excellent prospects for sustainability (long-term sponsoring and collaborations). Most of us are still eligible for individual talent grants like NWO Vidi and Vici and ERC Starting and Consolidator grants, providing an excellent additional basis for the longer-term sustainability and potential growth of the virtual institute.

Risk assessment

Risks	Mitigation
Language barrier in interdisciplinary collaboration (hurdles: differences in language, terminology, methodology, research culture, etc.).	Already demonstrated community-building workgroup, joint student projects, many PIs have inter-/interdisciplinary training, some have a dual appointment across institutions.
A large amount of talented tenure-track staff, drop out risk to other top institutions and/or companies.	Sustainable employability and development opportunities sponsored by host institutions.
Early career PIs lacking track record to pull large grants like NWO Gravity.	Build on other consortia over time with the involvement of senior personnel. Good prospects of meeting the eligibility criteria in the next 5-10 years.

2.d Visibility, attracting talents and new partners:

Attracting talents and engaging students and early career researchers

A distinct feature of this flagship program is that it comprises a new generation of young, high profile, internationally trained, diverse researchers (male-female ratio 50%-50%; 100% internationally trained; Dutch/non-Dutch 50%-50%). We are at a unique position to engage students and early career researchers because this consortium will be propelled by tenure track and young tenured professors. To further strengthen the link between our institutions within the consortium and with society, we will combine our networks and create a joint online presence (virtual institute website and social media presence) to attract new talent. The virtual institute of Integrative Neuromedicine will facilitate the coordination of hiring actions and promotion of projects to students.

We will create elective courses and summer schools that are uniquely tailored for current challenges (for example ethics of human enhancement; integrative digital twins; from lab to product) and that through one-to-one and (flipped) classroom interaction with students will capture their interest in integrative Neuromedicine; we will capitalize on this with our existing Bachelor and Masters programs of Neuroscience, Clinical Technology, Nanobiology etc., as well as the new minor "Integrative Neuromedicine" and Master program "Philosophy of Neuromedicine". Students will join the research conferences we will organize to present and learn. These conferences will also be used for contact with industrial partners.

Visibility

Flagship visibility will be enhanced in the scientific community by organizing workshops for international audiences, including talks streamed online, about the topics we work on; referral to the flagship in funding acknowledgments and affixing an affiliation to the virtual institute to our papers, websites etc. Importantly many of the PIs involved in our proposal have an established social media following on platforms like LinkedIn and Twitter. Visibility to stakeholders will be enhanced by outreach activities like workshops for patients (dealing with the concepts of (un)normality) for health care personnel (reflections on personalized intervention and ethics) and public. In fact, many of the PIs are already actively involved in Open Science activities in the area of public engagement, Citizens Science, Open Data and Open Access.

The ability to attract talents, partners and funding will also depend on our ability to promote the flagship/virtual institute as an entity, take part in strategic calls for new hires, and make our voice heard across the partner institutes when it comes to hiring decisions. This requires representation in department, university, national and European organizational entities. With members of the flagship represented in e.g. departmental counsels, NWO round tables, the Young Academy of Europe and associations of ERC grantees, we are well positioned to drive discussion at all levels. Our team collectively has an excellent track record that is internationally competitive. We are at the career stages that a large collection of open rank and career stage specific funding possibilities are available. As stated in our funding plan, we are confident that our scientific merits will assist us to successfully attract partners and external funding in the near future.

2.e Long term societal impact

Developing an Integrative Neuromedicine that includes the patient's perspective and ethical questions is of high relevance for tackling ND manifestation in current society. We will generate a comprehensive understanding of the brain (1), develop an integrative approach to neuromedicine (2), and connect efficient research, ethical reflection and clinical application (3), and establish an interdisciplinary training for students and early career researchers (4).

(1) We will generate a holistic picture of the brain that is useful for the general understanding of the developing and aging brain, enabling correlation of changes on different levels, and build a pipeline to increase the speed with which fundamental knowledge can be transferred to the clinic. Results will be made accessible via the virtual institute and website (2.d). Our project will serve as a best practice example for other integrative and interdisciplinary approaches in biomedicine.

Scientific impact: We will publish the scientific insights, making brain models available for different fields and applications through open access publication, sharing of datasets, the creation of online digital twins and the development of interventions to clinical trial level.

Valorization pathway: The developmental pipeline described in the proposal will stretch from TRL1-6. For clinical trials (TRL6) and subsequent market penetration we need an industrial network in the theranostics and pharma space, which will help valorization through further device and therapy developments. For example, in the context of the WP3 work on ultrasound-based intervention, a startup could be founded for which we will expand on our contacts in the Dutch and European investor landscape.

(2) We will make sure that unnecessary intervention, stigmatization and pathologization can be avoided. This will be supported by stimulating a broader societal debate on neurodiversity, good aging, standards of normality and ideals of optimality.

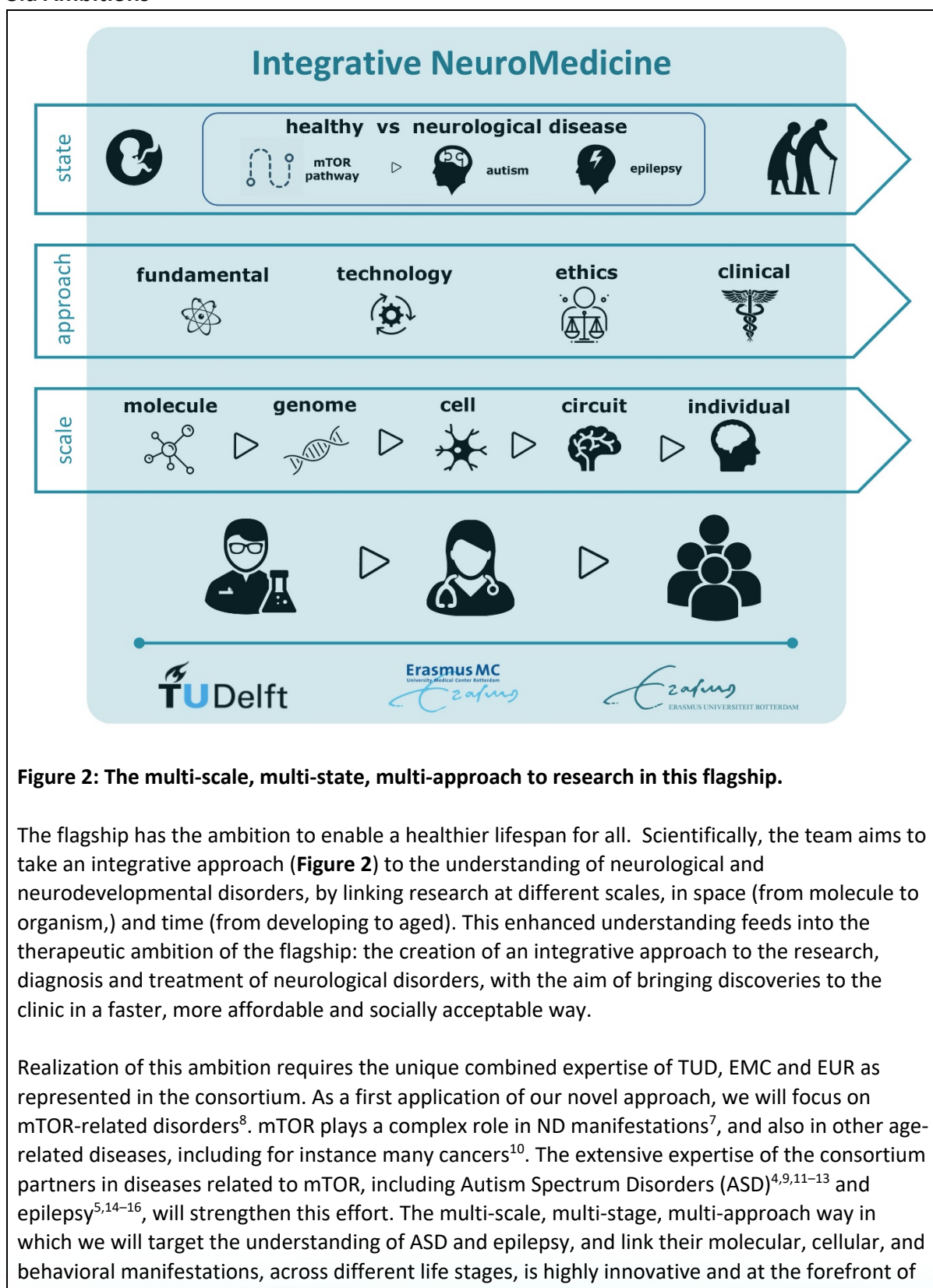
Societal impact: We will organize public debates and workshops for academics and layman audience, publishing in non-academic platforms for laymen, including art project / exhibition: 'A normal brain?' based on existing collaborations with CODARTS Rotterdam and the Rotterdam Arts and Science Lab (RASL) or through the Kavli Artist in Residence Program.

(3) We will facilitate tailor-made effective interventions based on partnership with patients, by mapping the ethical challenges and patients' concerns that must be addressed to increase the success, including individual and societal acceptance, of proposed interventions.

Healthcare impact: We will organize workshops with clinical staff and patients and generate health policy advice in collaboration with non-profit organizations and patient groups, and other stakeholders to lay the political and legal foundations for effective, individually tailored and affordable treatment and therapy.

(4) We will integrate the knowledge generated by the flagship in our educational programs. This fits very well with existing endeavors in for instance the nanobiology, molecular medicine, applied physics, life science and technology, neuroscience and philosophy programs.

Educational impact: We will develop an interdisciplinary Minor (short term) "Philosophy of Neuroscience" (based on existing Master courses and Bachelor of a Specific Discipline at the Erasmus School of Philosophy) and Master Program (long term) "Integrative Neuromedicine".

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d [1987 words]**3.a Ambitions****Figure 2: The multi-scale, multi-state, multi-approach to research in this flagship.**

The flagship has the ambition to enable a healthier lifespan for all. Scientifically, the team aims to take an integrative approach (**Figure 2**) to the understanding of neurological and neurodevelopmental disorders, by linking research at different scales, in space (from molecule to organism,) and time (from developing to aged). This enhanced understanding feeds into the therapeutic ambition of the flagship: the creation of an integrative approach to the research, diagnosis and treatment of neurological disorders, with the aim of bringing discoveries to the clinic in a faster, more affordable and socially acceptable way.

Realization of this ambition requires the unique combined expertise of TUD, EMC and EUR as represented in the consortium. As a first application of our novel approach, we will focus on mTOR-related disorders⁸. mTOR plays a complex role in ND manifestations⁷, and also in other age-related diseases, including for instance many cancers¹⁰. The extensive expertise of the consortium partners in diseases related to mTOR, including Autism Spectrum Disorders (ASD)^{4,9,11-13} and epilepsy^{5,14-16}, will strengthen this effort. The multi-scale, multi-stage, multi-approach way in which we will target the understanding of ASD and epilepsy, and link their molecular, cellular, and behavioral manifestations, across different life stages, is highly innovative and at the forefront of

the field. The challenge will lie in the coordination and translation of research efforts, which is why we define deeply cross-discipline WPs.

With this approach, this flagship contributes to the fundamentals of health and disease pillar of convergence, and to improved health journeys: we aim to both enhance our fundamental understanding of NDs and leverage this knowledge for increased efficiency therapies.

3.b Aims and feasibility

Aims:

WP1: Mapping: develop technological tools to study complementary brain models across life stages into old age.

WP2: Understanding: integrative disease modeling in complementary brain models and comparison to healthy models; reflection on normality vs. pathology; and integration of the obtained knowledge into healthy vs diseased digital twins.

WP3: Intervention: use insights from WP1 and 2 to create a framework for acceptable ethical intervention and translational pipelines for effective technological, chemical, biological and psychological interventions.

The general tools developed in these WPs will first be applied in studies of mTOR related disease to find new cross-scale and cross-stage correlations, identify actionable diagnostic markers, and develop novel, acceptable interventions.

Strategy to involve students:

All hired staff will serve as ambassadors between groups and institutions. Students motivated for the interdisciplinary research needed here (like those in the nanobiology program) will be jointly supervised between groups. Part of the funding will be used to reward joint master project proposals. A minor “Philosophy of Neuroscience” - based on Bachelor “Philosophy of a Specific Discipline” at ESPhil- and a joint master program “Integrative Neuromedicine”, taking elements from the nanobiology, neuroscience, computational neuroscience, philosophy and psychology master programs, will be created.

Feasibility and risk

Within the consortium we have all necessary knowledge to make the overarching aims of the flagship a reality.

Risk	Mitigation
Large scope of the project	-Start of a long-term endeavor in the form of a virtual institute is feasible because of young, high-potential applicants with long term commitment. -Focus during the initial period of the flagship on a single example: mTOR related disease, to get organization, culture and infrastructure in place.

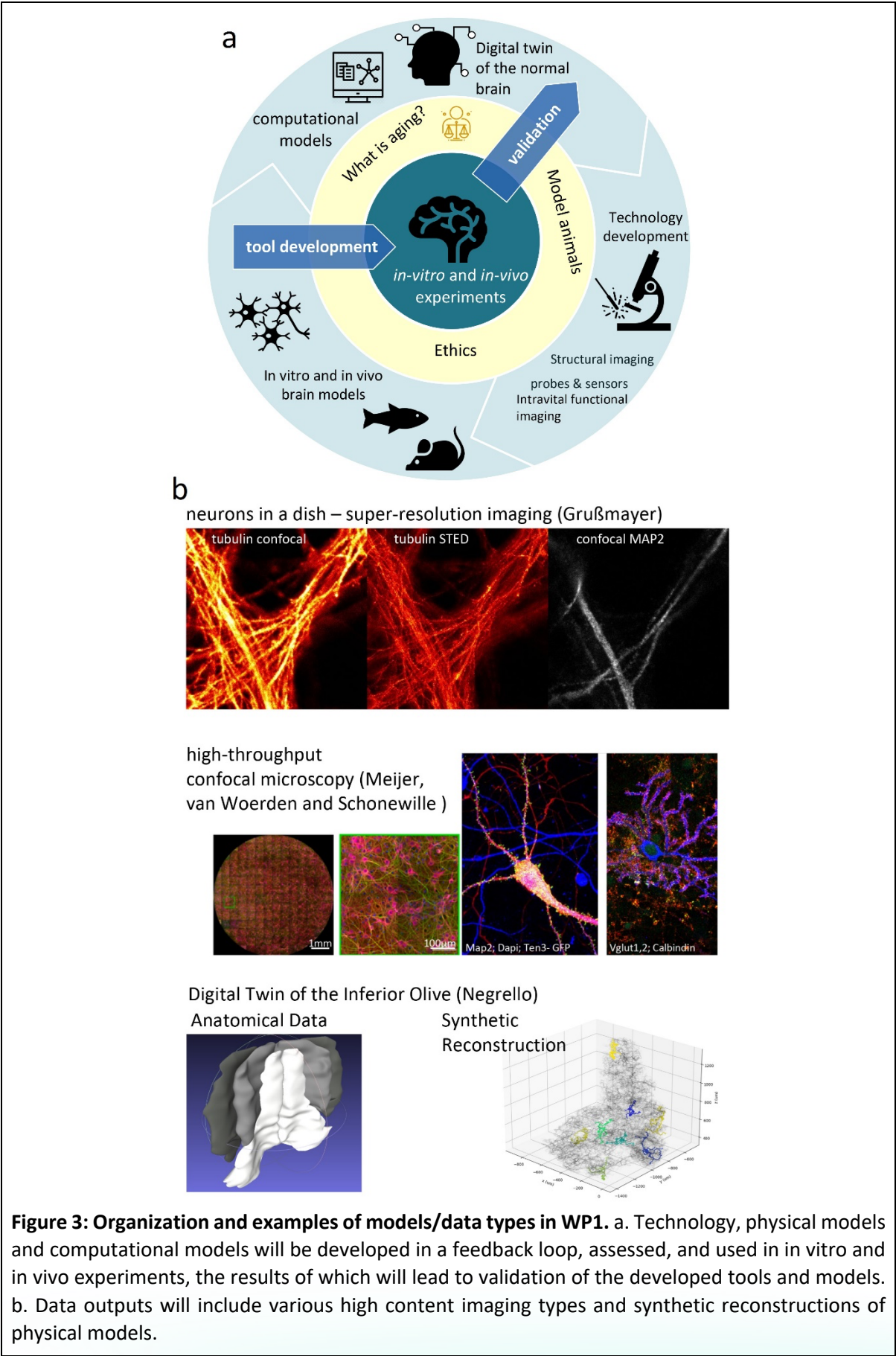
Knowledge transfer between WPs suboptimal	Staff will work across WPs (Figure 8, section 4b). The virtual institute infrastructure will also enable optimized knowledge transfer.
WP dependencies	WPs can start independent from each other but are highly integrated at later stages. For instance, WP2 researchers start on disease models, and integrate this at a later stage with the findings on the normal brain of WP1; similarly, WP3 researchers will start with fundamental diagnostic optimization, which can be specialized based on WP2 knowledge.

External funding

The consortium is made up of young, high potential researchers in all three institutions and matches perfectly with the Dutch national science agenda routes (Measuring and detection; NeurolabNL; Personalized medicine) and European programs (E-Brains). Development of diagnostic tools takes place in an environment in South-Holland rich with opportunities for public-private partnerships. We expect a combination of personal excellence grants, impact-based consortium grants and public-private funding to supplement our external funding.

3.c Work plan

WP1: Mapping focuses on concept and technology development; **WP2: Understanding** focuses on disease modeling, specifically of mTOR related diseases; and **WP3: Intervention** focuses on therapy development.



WP1: Mapping: To generate baseline data to define and treat neurodevelopmental and neurodegenerative diseases, we will establish different experimental and computational models as well as new microscopy technology to map molecular, cellular, and organizational network changes across spatial scales and stages of brain development (**Figure 3**).

1.1. Accessible in vitro experimental model to visualize neuronal circuits with molecular precision

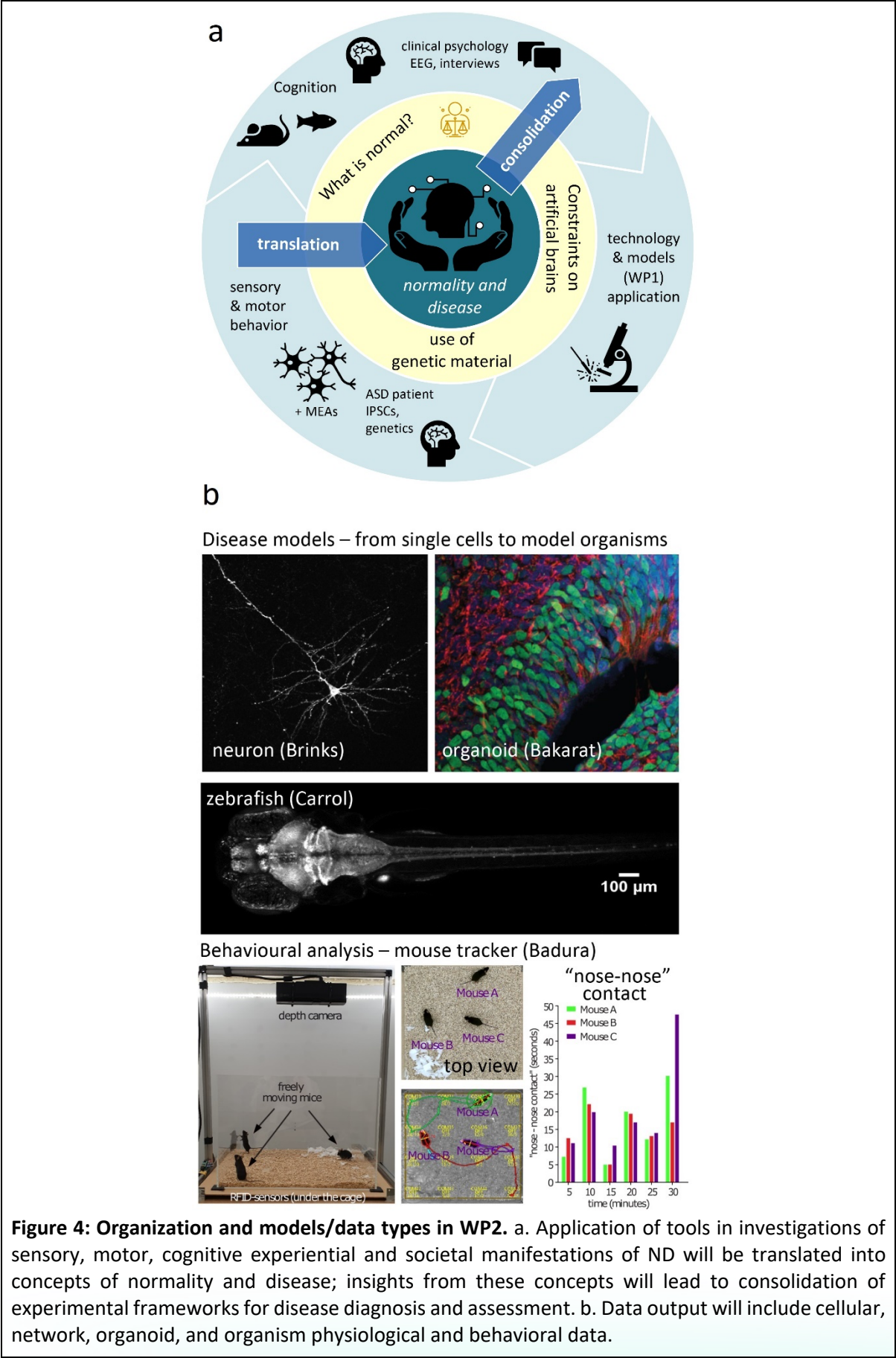
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|-------|---|
| 1.1.1 | Reconstitute complete neuronal in vitro 2D networks and 3D organoids from i) primary mouse hippocampal, cortical, and cerebellar neurons derived from embryonic, postnatal and adult mice ¹⁷⁻²⁰ , and ii) human iPSC derived neurons ^{21,22} . |
| 1.1.2 | Obtain high-resolution information on changes in network architecture and function by means of high-content automated confocal imaging correlated with super-resolution and quantitative phase imaging of morphology, connectivity and activity ^{18,23-28} . |

1.2. A workflow to study temporal dynamics of signaling in vivo

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|-------|---|
| 1.2.1 | Determine the generation of brain cell lineages through intravital imaging in model animals in genetically modified backgrounds ²⁹⁻³¹ . |
| 1.2.2 | Develop and implement advanced microscopy techniques, including calcium imaging, voltage imaging, light sheet microscopy, multiphoton imaging, and optogenetic perturbations in linking genetic backgrounds to brain activity patterns and behavioral outcomes ^{17,23,27,28,32-34} . |

1.3. A computational model of the changing brain

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|-------|---|
| 1.3.1 | Modelling molecular pathways with detailed molecular simulators (mCell and STEPS) and modeling of cell and network behavior in network simulators (neuroML + EDEN) ^{35,36} . |
| 1.3.2 | Describing neuronal homeostasis via connecting molecular pathways and network spiking models, in a digital twin of the experimental data ³⁷⁻³⁹ . |
| 1.3.3 | Discussing fundamental philosophical questions regarding the relation between the biological, computational, and experiential dimension (consciousness) of the brain ⁴⁰⁻⁴³ . |



WP2: Understanding. Understanding of brain functionality and effects of disease requires connecting experiential and behavioral data with network, cellular, electrophysiological and molecular data. In this WP, advanced measurement and computation techniques will be streamlined in pipelines for disease investigation, with Autism Spectrum Disorder (ASD) as an mTOR-related model (**Figure 4**).

2.1. Sensory and motor behavior: investigating phenotypes at the molecular, cellular and circuit level

2.1.1	Application of input-output assays based on advanced optical imaging (calcium, voltage), optogenetic actuators and multielectrode arrays on networks of neurotypical vs ASD patient cells to investigate molecular, cellular and circuit markers ^{23,24,28,44,45} .
2.1.2	Creation of comprehensive computational models (Digital twins) of 2D and 3D tissue models of ASD, and comparison to digital twins of healthy tissue models ^{35,37,38} .

2.2. Cognition: investigating phenotypes at the organism level

2.2.1	Investigation of behavioral markers of ASD in genetically modified mouse models ^{12,13,46,47} and linking to network level activity through intravital functional imaging (calcium, voltage) and optogenetic perturbations ^{24,48} .
2.2.2	Linking animal model behavior to patient experience in ASD through linking of genetic markers and behavior ^{9,13,22,49,50} .
2.2.3	Creation of network and cortical level computational models to explain the behavioral data and provide the possibility of prediction and regulation ^{38,39,51} .

2.3. Ethical implications: investigating ASD dysfunction and normality at the societal level

2.3.1	Conceptual analysis: theoretical and qualitative research on objective and lived health, the concept of normality and its normative implications, including evaluation of existing interviews with patients ^{52,53} .
2.3.2	Integrating qualitative results and patient experience into the design of digital twins to bridge lived and objectified health/normality ^{40,42,43,54} .

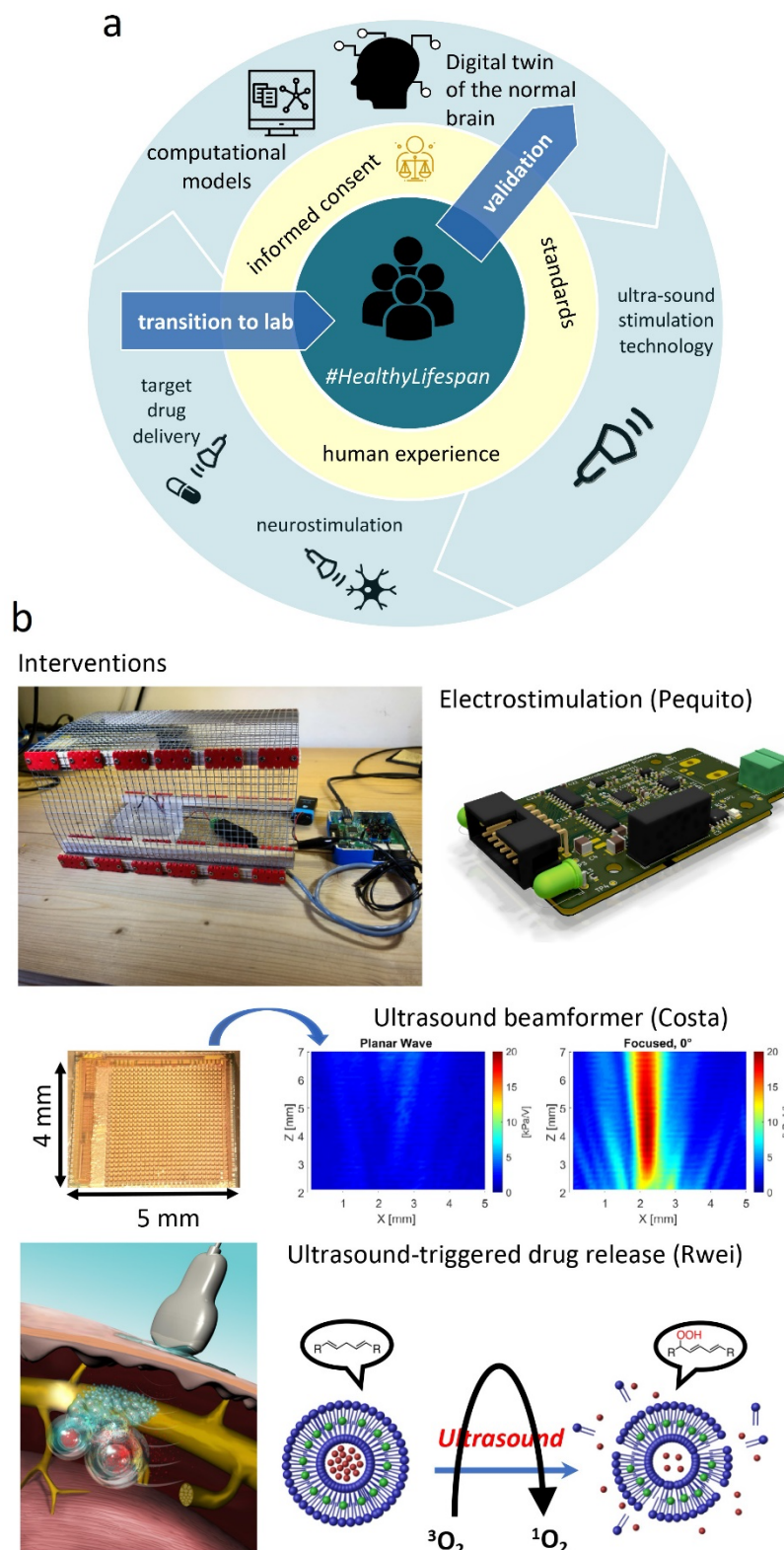


Figure 5: Organization and tools in WP3. a. Digital twins and developed therapeutic interventions, focused on neuromodulation and drug delivery, will provide information on effective therapeutic routes, which will be ethically and philosophically assessed and inform the disease modeling and fundamental work across life stages and scales in WP1 and 2; insights from the lab will in turn be validated in this WP. b. Techniques used include electrostimulation, ultrasound stimulation and ultrasound-triggered drug-release.

WP3: Intervention. By converging the perspectives of engineers, clinicians, and ethicists during the research stage, we aim to improve the translational pipeline for therapies. In this WP, we will develop, and investigate the application of, advanced drug delivery systems and neuromodulation therapies. We will utilize fundamental electrophysiological, optogenetic, and ultrasound data and make use of digital twins for generalization and affordability (**Figure 5**).

3.1. Mechanical, optical, and electrical detection and intervention for mTOR-related disease

3.1.1	Development of ultrasound neuromodulation microchips for mTOR pathway-related diseases (e.g., epilepsy) ^{37,55} ; inferring their neuromodulation capabilities with known disease models and electrophysiological readouts (such as EEG) ^{39,56–58} .
3.1.2	Testing of neuromodulation on model animals and tissue models ^{11,13,14,47,59} , linking mTOR pathway disease (i.e. epilepsy) to brain development and investigating preventative intervention ^{60,61} .
3.1.3	Testing of neuromodulation and mTOR-related disease effects on exposure therapy in anxiety disorders ^{62–65} , linking molecular markers and intervention choices to patient wellbeing.

3.2 Advanced drug delivery systems (e.g., externally-triggerable) to pharmaceutically target the mTOR pathway via gene and/or small molecule drugs

3.2.1	Engineer ultrasound-triggerable drug delivery systems whose spatiotemporal drug release profile can be controlled through ultrasound ^{66,67} .
3.2.2	Utilize the insights developed in WP1 and WP2 to optimize the drug molecule species, timing, and dosage required for therapeutic interventions.

3.3 Implementation of intervention techniques

3.3.1	Computational modeling of the mechanisms of intervention for evaluation of intervention effects, focusing on improvements in epileptic seizures and mitigation of cognitive deficits related to the mTOR pathway, building on insights from WP2.
3.3.2	Building on insights from WP2.3, investigate ethical challenges of intervention in ND manifestation (concerning identity, autonomy, authenticity), in an integrative partnership approach, considering the patient's perspective.

Required facilities

- Mouse and zebrafish facilities in Delft and Rotterdam (present);
- High performance computing facilities in Delft and Rotterdam (present);
- ML1 and ML2 tissue culturing in Delft and Rotterdam (present);
- DM1 and DM2 Experimental rooms for optical and ultrasound measurements (present).

Data-and infrastructure plan

Shared data-access for consortium members is necessary. Institutional servers will be the primary storage facilities. Published data will be open access stored on the 4TU repository and EUR "Pure"

repository. Shared access for consortium members will initially be relatively straightforwardly arranged through approved Teams and SharePoint repositories, awaiting a tailored convergence solution. Data management plans will be approved by institutional data-stewards.

3.d Alignment of the program with relevant national and international research agendas

The flagship aligns with at least 4 routes of the Dutch national science agenda (Health care research, Sickness prevention and treatment; Measuring and detection; NeurolabNL; Personalized medicine), the Hoofdzaken initiative, and would fit well with the renewal of the European Union endeavor for computational understanding of the brain (Human Brain Project; E-Brains). Moreover, it could make us a valuable partner in the major US based NSF initiative Brain Research through Advancing Innovative Neurotechnologies (BRAIN). The high-risk, high-gain objective is a completely new, integrative way to develop understanding and therapies for neurodevelopmental/neurodegenerative diseases; the specific focus on mTOR fits well within this scope as this pathway can affect brain function throughout life and affects a large disease population, because of which an integrative approach is needed to advance understanding and interventions. Creating an efficient pathway to the clinic for therapies for mTOR-related NDs, making use of streamlined knowledge and digital twins for generalization and affordability will therefore have a large impact on science, medicine and society.

4. Consortium (25%) max 2000 words for 4.a to 4.d [1405 words]

4.a Track record of the consortium members:

Track record in academic research, translation of scientific results into clinical application or societal impact

We represent a broad spectrum of talent in “cutting-edge areas” (**Figure 7**). Consortium members have background and training ranging from natural sciences to medicine to the humanities and we work on and bridge different disciplines covering fundamentals, technology development, ethics and clinical research. Our excellent track records and the impact our research has already made are reflected in:

Academic research:

-Landmark publications in relevant journals for our disciplines across all areas of expertise^{19-21,23,24,28,31,33,34,41,42,45,58,68-76}.

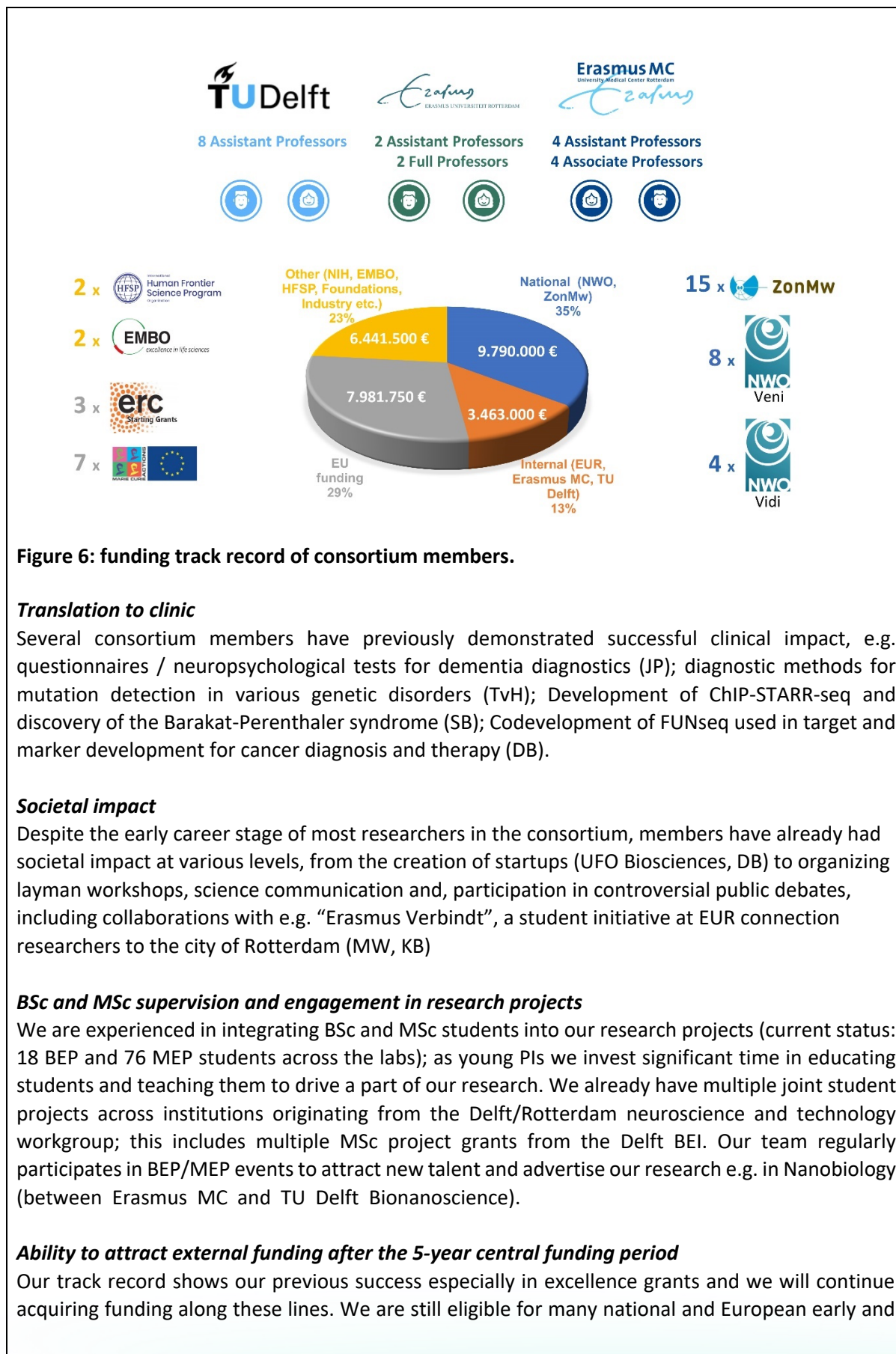
-A significant number of personal excellence grants as postdocs (e.g. Marie Skłodowska-Curie Actions (DM, MJ), Rubicon (DB), NWO Veni (SB, DM, MS, TvH, AB, IF, ZG, GvW))

- A significant number of personal excellence grants as independent group leaders (e.g. VW stiftung (MW), NWO Vidi/StartUp (DB, AB, IF, ZG, GvW), ERC StG (DB, ZG,MS), ERC PoC (MN), DFG (KB, MS, MW), sectorplan(KG), Erasmus MC/EUR fellowships (SB, MS, AB, ZG, GvW)), RPI (SP), HFSP (SB,TC), NIH (AR);

- Involvement in relevant internal and external consortia (e.g. TU Delft AI initiative BIOLab (DB,KG, EC), HBP (MN), MSCA-ITN (AB), ZonMW (EC, TvH, SB, IF, JP))

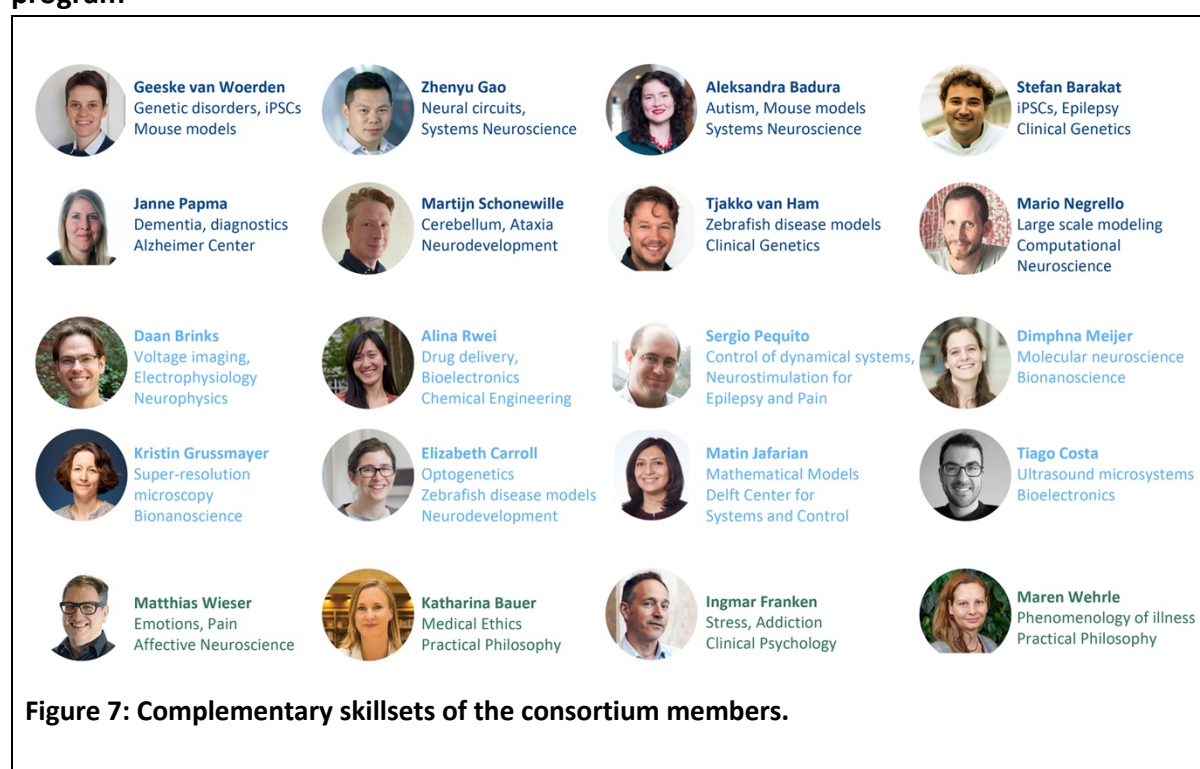
- Awards like the KNAW Early Career Award (SB) or the Young Outstanding Researcher award from Alzheimer Nederland (JP).

-Despite our young career as PIs (<4 years on average), we already succeeded in bringing in a total of €18864250 of internal, national, European and other funding (**Figure 6**).



mid-career awards. With the added credibility and strength through this flagship program, we will have the critical mass and spectrum of expertise to be able to apply for larger consortia grants; see funding plans 3b, alignment with national/European research agendas 3d and long-term sustainability plan 2c. We expect that, through positioning ourselves as a unified virtual institute, and using the results of the flagship, we will also be able to attract targeted private sponsoring. With this goal in mind, we will actively promote the Integrative Neuromedicine Institute in contacts with private parties, making use of our existing relationships with business developers from TUD, EMC, EUR and Oncode institute to create brand recognition in the health tech, diagnostics and pharmaceutical industries.

4.b Expertise of the consortium members in relation to the aims and objectives of the program



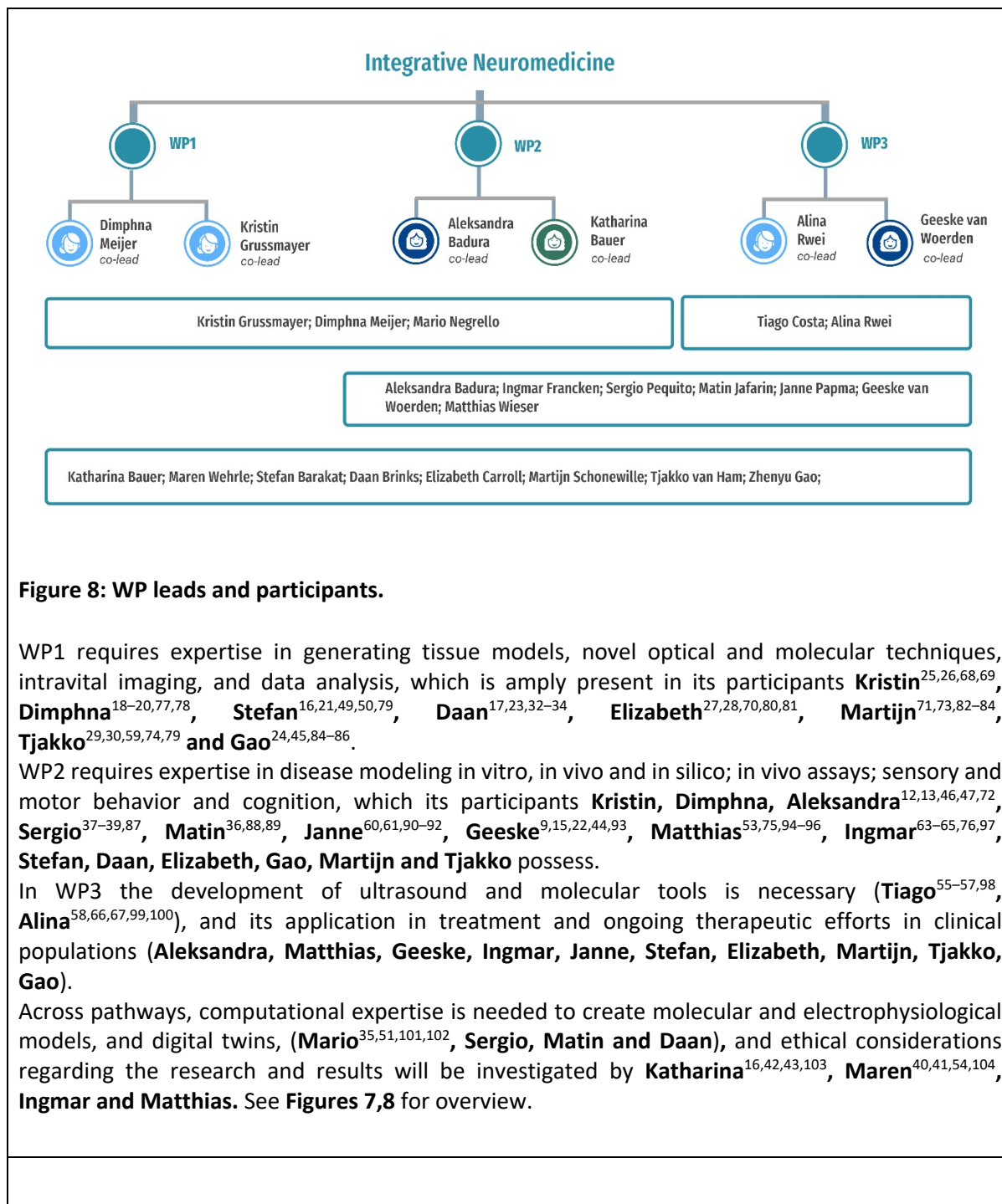


Figure 8: WP leads and participants.

WP1 requires expertise in generating tissue models, novel optical and molecular techniques, intravital imaging, and data analysis, which is amply present in its participants **Kristin**^{25,26,68,69}, **Dimphna**^{18–20,77,78}, **Stefan**^{16,21,49,50,79}, **Daan**^{17,23,32–34}, **Elizabeth**^{27,28,70,80,81}, **Martijn**^{71,73,82–84}, **Tjakko**^{29,30,59,74,79} and **Gao**^{24,45,84–86}.

WP2 requires expertise in disease modeling in vitro, in vivo and in silico; in vivo assays; sensory and motor behavior and cognition, which its participants **Kristin**, **Dimphna**, **Aleksandra**^{12,13,46,47,72}, **Sergio**^{37–39,87}, **Matin**^{36,88,89}, **Janne**^{60,61,90–92}, **Geeske**^{9,15,22,44,93}, **Matthias**^{53,75,94–96}, **Ingmar**^{63–65,76,97}, **Stefan**, **Daan**, **Elizabeth**, **Gao**, **Martijn** and **Tjakko** possess.

In WP3 the development of ultrasound and molecular tools is necessary (**Tiago**^{55–57,98}, **Alina**^{58,66,67,99,100}), and its application in treatment and ongoing therapeutic efforts in clinical populations (**Aleksandra**, **Matthias**, **Geeske**, **Ingmar**, **Janne**, **Stefan**, **Elizabeth**, **Martijn**, **Tjakko**, **Gao**).

Across pathways, computational expertise is needed to create molecular and electrophysiological models, and digital twins, (**Mario**^{35,51,101,102}, **Sergio**, **Matin** and **Daan**), and ethical considerations regarding the research and results will be investigated by **Katharina**^{16,42,43,103}, **Maren**^{40,41,54,104}, **Ingmar** and **Matthias**. See **Figures 7,8** for overview.

4.c Involvement of partners:

Partners include scientific partners external to the consortium, industrial partners and other stakeholders.

The consortium members are well placed in broader scientific networks, representing 11 departments across 3 universities and including members of for instance the young academy of Europe, the association of ERC grantees, Medical Delta, Delft Health initiative and Delft AI labs. This ensures that a broad palette of collaborators is available when needed.

The consortium members are driven by an urge towards understanding disease and clinical application of their work, but nevertheless have good connections with industrial partners including Single Quantum; UFO Biosciences; Delmic BV; ThermofisherScientific; NeCEN; Quantib; Neurasmus, Pharming Global and Imec. These partners mostly contribute on the technology development side at the level of individual labs. We will involve them at an early stage in our outreach and dissemination to achieve partnering at the level of the Integrative Neuromedicine Institute.

Our primary other stakeholders are patients. Interacting with patients and patient organizations is an explicit part of the ethical investigations part of the proposal; they will be involved throughout the lifetime of this project. Patient organizations will be part of the advisory board of the virtual institute.

Given the stated goal of a healthier lifespan for all, expedited translation of scientific results to the clinic requires expertise on the legal, financial and insurance aspects and will be sought throughout the project.

4.d Governance, organization and collaboration strategy of the consortium

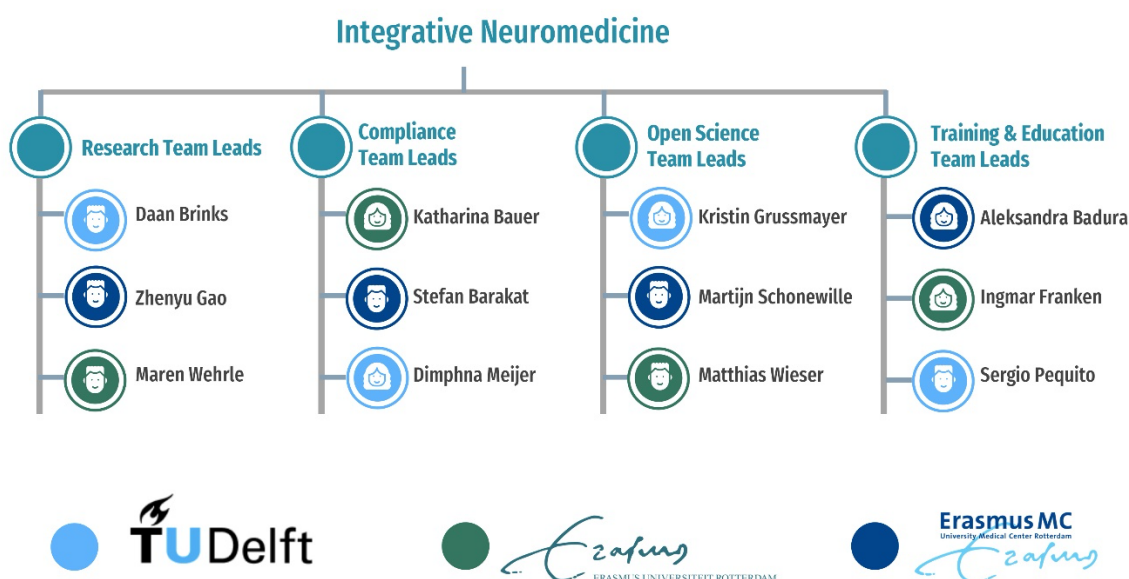


Figure 9: leadership structure of the consortium.

Governance, organization and collaboration are facilitated as sketched in **Figures 8 and 9**.

We believe in a democratic decision-making process and view this flagship as a virtual institute in which we are all PIs. This means an expectation of service and a division of responsibility for various key aspects of organization in committees. There will be regular faculty meetings (once per 2 months) in which budgetary considerations, project deliverables and planning for project progress will be shared and discussed with the faculty. The institute leads are ultimately responsible for setting the agenda of these meetings, but committees focusing on research progress (WP leads), compliance, open science and education will be formed and report back to the faculty meeting (**Figure 9**).

An advisory board consisting of international experts, stakeholders such as patient organizations, legal and valorization experts will be put in place; they will meet yearly with the institutional leads.

Collaboration is facilitated by the fact that convergence personnel will work between groups from different institutes and across WPs (**Figure 8**), which will lead to shared project meetings and group meetings. We already have yearly science days in the neuroscience and technology workgroup that this flagship originated in; those will be used for a day of talks and poster presentations by PIs, postdocs and students. Industrial partners will be invited to these meetings as well to facilitate novel collaborations. We will fund shared master projects, in a model similar to the Delft Bioengineering institute that we have experience in.

News regarding the flagship will be shared in a virtual institute newsletter, which will contain information on new personnel joining the virtual institute, papers published, and funding opportunities.

We expect collaboration will grow stronger when the identity of the virtual institute is strengthened. Starting from the beginning of the flagship project, internal and external funding will be sought for strengthening of the organizational infrastructure of the institute, i.e. administrative assistance, dedicated business development etc. we will use as our models the Delft Bioengineering institute, Delft AI labs and Oncode, all virtual institutes with strong identities that we participate in or can learn from.

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table):

A pluriannual plan will be required if your proposal is accepted

II Letters of intent and letters of commitment:

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

III Letter of support from department head of the main lead:

On headed paper of the department or institution.

IV References

Annex I Budget table (including the external funding table):

A pluriannual plan will be required if your proposal is accepted

Budget formulier: Convergentie Flagship Call

Flagship:	Integrative Neuromedicine
Flagship acronym:	IN
Naam Penvoerder:	Daan Brinks
Startdatum en looptijd van het project:	01-09-2022, 5 jaar
Totaal project budget:	11.073.585 Euro

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1.	Loonkosten:								2022		2023		2024		2025		2026		Totaal		Totaal per partner
	Partner die de kosten maakt	Functie	Aantal fte	Startdatum	Einddatum	Inschaling	Maandbedrag	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
	Onderzoeksorganisaties																				
	TU Delft/Faculty of AS/Dept of imaging physics	postdoc	1.0	9/1/2022	8/31/2025	Schaal 10	€ 4,181	€ 225,774	€ 72,543.00		€ 75,206		€ 78,025						€ 225,774	€ 0	€ 225,774.00
	TU Delft/Faculty of AS/Dept of Bionanoscience	PhD Student	1.0	9/1/2022	8/31/2026	P_TUD	€ 3,101	€ 223,272	€ 52,819.00		€ 55,460		€ 57,401		€ 57,592				€ 223,272	€ 0	€ 223,272.00
	TU Delft/Faculty of AS/Dept of Chemical Engineering	postdoc	1.0	1/1/2023	12/31/2025	Schaal 10	€ 4,181	€ 225,774	€ 72,543.00		€ 75,206		€ 78,025						€ 225,774	€ 0	€ 225,774.00
	Erasmus Medisch Centrum/ Dept of Neuroscience	PhD Student	1.0	9/1/2022	8/31/2026	P_EMC	€ 3,101	€ 223,272	€ 52,819.00		€ 55,460		€ 57,401		€ 57,592				€ 223,272	€ 0	€ 223,272.00
	Erasmus Medisch Centrum/ Dept of Neuroscience	postdoc	1.0	9/1/2023	8/31/2026	Schaal 10	€ 4,181	€ 225,774			€ 72,543.00		€ 75,206		€ 78,025				€ 225,774	€ 0	€ 225,774.00
	Erasmus Medisch Centrum/ Dept of Neuroscience	postdoc	1.0	1/1/2024	12/31/2026	Schaal 10	€ 4,181	€ 225,774					€ 72,543.00		€ 75,206		€ 78,025		€ 225,774	€ 0	€ 225,774.00
	Erasmus Universiteit Rotterdam/Dept of Philosophy	PhD Student	1.0	9/1/2022	8/31/2026	P_EUR	€ 3,518	€ 253,296	€ 51,991.00		€ 62,615		€ 66,793.00		€ 71,897				€ 253,296	€ 0	€ 253,296.00
	TU Delft/Faculty of applied sciences/Dept of imaging physics	PhD Student	1.0	1/1/2022	12/31/2023	P_TUD	€ 3,187	€ 114,732	€ 57,364.00		€ 57,368								€ 114,732	€ 0	€ 114,732.00
	TU Delft/Faculty of applied sciences/Dept of imaging physics	PhD Student	1.0	1/1/2022	12/31/2024	P_TUD	€ 3,187	€ 172,098	€ 57,362.00		€ 57,368		€ 57,368.00						€ 172,098	€ 0	€ 172,098.00
	TU Delft/Faculty of applied sciences/Dept of imaging physics	PhD Student	1.0	1/1/2022	12/31/2024	P_TUD	€ 3,187	€ 172,098	€ 57,362.00		€ 57,368		€ 57,368.00						€ 172,098	€ 0	€ 172,098.00
	TU Delft/Faculty of applied sciences/Dept of imaging physics	postdoc	1.0	1/1/2023	12/31/2025	Schaal 10	€ 4,444	€ 239,976			€ 79,992		€ 79,992.00		€ 79,992				€ 239,976	€ 0	€ 239,976.00
	TU Delft/Faculty of applied sciences/Dept of imaging physics	postdoc	1.0	1/1/2024	12/31/2026	Schaal 10	€ 4,444	€ 239,976					€ 79,992		€ 79,992.00		€ 79,992		€ 239,976	€ 0	€ 239,976.00
	TU Delft/Faculty of applied sciences/Dept of imaging physics	postdoc	1.0	7/1/2022	12/31/2023	Schaal 10	€ 4,444	€ 119,988	€ 39,996.00		€ 79,992								€ 119,988	€ 0	€ 119,988.00
	TU Delft/Faculty of applied sciences/Dept of imaging physics	PhD Student	1.0	1/1/2023	12/31/2026	P_TUD	€ 3,187	€ 229,464			€ 57,360.00		€ 57,368		€ 57,368		€ 57,368		€ 229,464	€ 0	€ 229,464.00
	TU Delft/Faculty of applied sciences/Dept of Bionanoscience	PhD Student	0.4	1/1/2022	12/31/2025	P_TUD	€ 3,187	€ 80,312	€ 20,078.00		€ 20,078.00		€ 20,078		€ 20,078				€ 80,312	€ 0	€ 80,312.00
	TU Delft/Faculty of applied sciences/Dept of Bionanoscience	PhD Student	0.4	9/1/2022	8/31/2026	P_TUD	€ 3,187	€ 80,312	€ 20,078.00		€ 20,078.00		€ 20,078.00		€ 20,078				€ 80,312	€ 0	€ 80,312.00
	TU Delft/Faculty of applied sciences/Dept of Bionanoscience	PhD Student	0.4	9/1/2023	8/31/2027	P_TUD	€ 3,187	€ 80,312			€ 20,078.00		€ 20,078.00		€ 20,078		€ 20,078		€ 80,312	€ 0	€ 80,312.00
	TU Delft/Faculty of applied sciences/Dept of Bionanoscience	PhD Student	0.4	1/1/2022	12/31/2024	P_TUD	€ 3,187	€ 60,234	€ 20,078.00		€ 20,078.00		€ 20,078						€ 60,234	€ 0	€ 60,234.00
	TU Delft/Faculty of applied sciences/Dept of Bionanoscience	PhD Student	0.4	9/1/2022	8/31/2025	P_TUD	€ 3,187	€ 60,234			€ 20,078.00		€ 20,078.00		€ 20,078.00				€ 60,234	€ 0	€ 60,234.00
	TU Delft/Faculty of applied sciences/Dept of Bionanoscience	PhD Student	0.4	9/1/2022	8/31/2025	P_TUD	€ 3,187	€ 60,234			€ 20,078.00		€ 20,078.00		€ 20,078				€ 60,234	€ 0	€ 60,234.00
	TU Delft/Faculty of 3ME/Delft center for systems and control	PhD Student	1.0	1/1/2023	6/1/2024	P_TUD	€ 3,187	€ 86,049			€ 57,365.00		€ 28,684.00						€ 86,049	€ 0	€ 86,049.00
	TU Delft/Faculty of 3ME/Delft center for systems and control	PhD Student	0.5	1/1/2022	12/31/2022	P_TUD	€ 3,187	€ 28,683	€ 28,683.00										€ 28,683	€ 0	€ 28,683.00
	TU Delft/Faculty of 3ME/Delft center for systems and control	PhD Student	1.0	1/1/2023	12/31/2026	P_TUD	€ 3,187	€ 229,464			€ 57,360.00		€ 57,368		€ 57,368		€ 57,368		€ 229,464	€ 0	€ 229,464.00
	TU Delft/Faculty of 3ME/Delft center for systems and control	PhD Student	1.0	1/1/2023	12/31/2026	P_TUD	€ 3,187	€ 229,464			€ 57,360.00		€ 57,368		€ 57,368		€ 57,368		€ 229,464	€ 0	€ 229,464.00
	TU Delft/Faculty of 3ME/Delft center for systems and control	PhD Student	1.0	1/1/2024	12/31/2026	P_TUD	€ 3,187	€ 172,098					€ 57,362		€ 57,368		€ 57,368		€ 172,098	€ 0	€ 172,098.00
	TU Delft/Faculty of 3ME/Delft center for systems and control	PhD Student	1.0	1/1/2025	12/31/2026	P_TUD	€ 3,187	€ 114,732							€ 57,364		€ 57,368		€ 114,732	€ 0	€ 114,732.00
	TU Delft/Faculty of 3ME/Delft center for systems and control	PhD Student	0.2	7/1/2022	6/30/2026	P_TUD	€ 3,187	€ 50,023	€ 10,023.00		€ 10,000.00		€ 10,000		€ 10,000		€ 10,000		€ 50,023	€ 0	€ 50,023.00
	TU Delft/Faculty of 3ME/Delft center for systems and control	PhD Student	0.4	7/1/2022	6/30/2026	P_TUD	€ 3,187	€ 100,000	€ 10,000.00		€ 20,000.00		€ 30,000		€ 30,000		€ 10,000		€ 100,000	€ 0	€ 100,000.00
	TU Delft/Faculty of 3ME/Delft center for systems and control	PhD Student	0.9	1/1/2023	12/31/2026	P_TUD	€ 3,187	€ 200,001			€ 50,001.00		€ 50,000		€ 50,000		€ 50,000		€ 200,001	€ 0	€ 200,001.00
	TU Delft/Faculty of EEMCS/Dept of microelectronics	PhD Student	1.0	1/1/2022	12/31/2023	P_TUD	€ 3,187	€ 114,732	€ 57,364.00		€ 57,368								€ 114,732	€ 0	€ 114,732.00
	TU Delft/Faculty of EEMCS/Dept of microelectronics	PhD Student	1.0	4/1/2022	6/1/2024	P_TUD	€ 3,187	€ 129,074	€ 57,364.00		€ 57,368		€ 14,342						€ 129,074	€ 0	€ 129,074.00
	TU Delft/Faculty of EEMCS/Dept of microelectronics	PhD Student	1.0	4/1/2022	3/31/2026	P_TUD	€ 3,187	€ 229,464	€ 43,018.00		€ 57,368		€ 57,368		€ 57,368		€ 14,342		€ 229,464	€ 0	€ 229,464.00
	TU Delft/Faculty of EEMCS/Dept of microelectronics	PhD Student	1.0	1/1/2023	12/31/2026	P_TUD	€ 3,187	€ 229,464			€ 57,360		€ 57,368		€ 57,368		€ 57,368		€ 229,464	€ 0	€ 229,464.00
	TU Delft/Faculty of EEMCS/Dept of microelectronics	PhD Student	1.0	9/1/2022	8/31/2026	P_TUD	€ 3,187	€ 229,464	€ 14,334.00		€ 57,368		€ 57,368		€ 57,368		€ 43,026		€ 229,464	€ 0	€ 229,464.00
	TU Delft/Faculty of EEMCS/Dept of microelectronics	PhD Student	1.0	9/1/2022	8/31/2026	P_TUD	€ 3,187	€ 229,464	€ 14,334.00		€ 57,368		€ 57,368		€ 57,368		€ 43,026		€ 229,464	€ 0	€ 229,464.00
	TU Delft/Faculty of EEMCS/Dept of microelectronics	postdoc	1.0	9/1/2022	9/1/2025	Schaal 10	€ 4,095	€ 227,273			€ 75,756		€ 75,756		€ 75,761				€ 227,273	€ 0	€ 227,273.00

TU Delft/Faculty of AS/Dept of Chemical Engineering	postdoc	1.0	1/1/2022	12/31/2023	Schaal 10	€	4,209	€ 151,524	€ 75,756.00	€ 75,768						€ 151,524	€ 0	€ 151,524.00
TU Delft/Faculty of AS/Dept of Chemical Engineering	PhD Student	1.0	1/1/2022	5/31/2023	P_TUD	€	3,187	€ 81,269	€ 57,368.00	€ 23,901						€ 81,269	€ 0	€ 81,269.00
TU Delft/Faculty of AS/Dept of Chemical Engineering	PhD Student	1.0	1/1/2024	12/31/2026	P_TUD	€	3,187	€ 172,098			€ 57,362		€ 57,368	€ 57,368		€ 172,098	€ 0	€ 172,098.00
Erasmus Medisch Centrum/ Dept of Neuroscience	PhD Student	1.0	1/1/2022	12/31/2024	P_EMC	€	3,187	€ 172,098	€ 57,362.00	€ 57,368	€ 57,368					€ 172,098	€ 0	€ 172,098.00
Erasmus Medisch Centrum/ Dept of Neuroscience	postdoc	1.0	1/1/2023	12/31/2026	Schaal 10	€	3,889	€ 280,008		€ 70,000	€ 70,000		€ 70,000	€ 70,008		€ 280,008	€ 0	€ 280,008.00
Erasmus Medisch Centrum/ Dept of Neuroscience	PhD Student	1.0	1/1/2023	12/31/2026	P_EMC	€	3,187	€ 229,464		€ 57,360	€ 57,368		€ 57,368	€ 57,368		€ 229,464	€ 0	€ 229,464.00
Erasmus Medisch Centrum/ Dept of Neuroscience	PhD Student	1.0	1/1/2022	12/31/2023	P_EMC	€	3,187	€ 114,732	€ 57,364.00	€ 57,368						€ 114,732	€ 0	€ 114,732.00
Erasmus Medisch Centrum/ Dept of Neuroscience	postdoc	1.0	1/1/2022	12/31/2023	Schaal 10	€	4,209	€ 151,524	€ 75,756.00	€ 75,768						€ 151,524	€ 0	€ 151,524.00
Erasmus Medisch Centrum/ Dept of Neuroscience	postdoc	1.0	1/1/2023	12/31/2026	Schaal 10	€	4,209	€ 303,048		€ 75,756	€ 75,756		€ 75,756	€ 75,780		€ 303,048	€ 0	€ 303,048.00
Erasmus Medisch Centrum/ Dept of Neuroscience	PhD Student	1.0	1/1/2023	12/31/2026	P_EMC	€	3,187	€ 229,464		€ 57,360	€ 57,368		€ 57,368	€ 57,368		€ 229,464	€ 0	€ 229,464.00
Erasmus Medisch Centrum/ Dept of Neuroscience	PhD Student	1.0	2/1/2022	1/31/2026	P_EMC	€	3,187	€ 229,464	€ 57,360	€ 57,368	€ 57,368		€ 57,368			€ 229,464	€ 0	€ 229,464.00
Erasmus Medisch Centrum/ Dept of Neuroscience	PhD Student	1.0	1/1/2022	12/31/2022	P_EMC	€	3,187	€ 57,366	€ 57,366							€ 57,366	€ 0	€ 57,366.00
Erasmus Medisch Centrum/ Dept of Neuroscience	postdoc	1.0	1/1/2023	12/31/2024	Schaal 10	€	4,209	€ 151,524		€ 75,756.00	€ 75,768					€ 151,524	€ 0	€ 151,524.00
Erasmus Medisch Centrum/ Dept of Neuroscience	PhD Student	1.0	1/1/2023	12/31/2025	P_EMC	€	3,187	€ 172,098		€ 57,362.00	€ 57,368		€ 57,368			€ 172,098	€ 0	€ 172,098.00
Erasmus Medisch Centrum/ Dept of Neuroscience	PhD Student	1.0	3/1/2022	2/28/2026	P_EMC	€	3,187	€ 229,464	€ 39,992.00	€ 57,368.00	€ 57,368		€ 57,368	€ 17,368		€ 229,464	€ 0	€ 229,464.00
Erasmus Medisch Centrum/ Dept of Neuroscience	postdoc	1.0	1/1/2024	12/31/2026	Schaal 10	€	4,209	€ 227,286			€ 75,756		€ 75,756	€ 75,774		€ 227,286	€ 0	€ 227,286.00
Erasmus Medisch Centrum/ Dept of Neuroscience	postdoc	1.0	1/1/2024	12/31/2026	Schaal 10	€	4,209	€ 227,286			€ 75,756		€ 75,756	€ 75,774		€ 227,286	€ 0	€ 227,286.00
Erasmus Medisch Centrum/ Dept of Neuroscience	PhD Student	1.0	1/1/2022	12/31/2023	P_EMC	€	3,187	€ 114,732	€ 57,364	€ 57,368						€ 114,732	€ 0	€ 114,732.00
Erasmus Medisch Centrum/ Dept of Neuroscience	Tenure tracker	1.0	1/1/2022	6/30/2023	Schaal 11	€	4,639	€ 125,253	€ 61,697	€ 63,556						€ 125,253	€ 0	€ 125,253.00
Erasmus Universiteit Rotterdam/Dept of Clinical Psychology	PhD Student	1.0	9/1/2022	8/31/2026	P_EUR	€	3,187	€ 229,464		€ 57,360	€ 57,368		€ 57,368	€ 57,368		€ 229,464	€ 0	€ 229,464.00
Ondernemingen																		€ 0.00
Onderneming 2						€	-	€ 0										€ 0.00
Onderneming 3						€	-	€ 0										€ 0.00
Onderneming 4						€	-	€ 0										€ 0.00
Overige (private) partners																		€ 0.00
Naam 1						€	-	€ 0								€ 0	€ 0	€ 0.00
Naam 2						€	-	€ 0								€ 0	€ 0	€ 0.00
Naam 3						€	-	€ 0								€ 0	€ 0	€ 0.00
Naam 4						€	-	€ 0								€ 0	€ 0	€ 0.00
Totaal								€ 9,561,522								Totaal loonkosten		€ 9,561,521

2.

Kosten van materialen en hulpmiddelen:				2022		2023		2024		2025		2026		Totaal		Totaal per partner
Partner die de kosten maakt	Materiaal / hulpmiddel		Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																
TU Delft/Faculty of applied sciences/Dept of imaging physics	consumables postdoc WP 1-2		€ 39,940	€ 19,940.00		€ 10,000		€ 10,000.00						€ 39,940	€ 0	€ 39,940.00
TU Delft/Faculty of applied sciences/Dept of Bionanoscience	consumables PhD student WP1		€ 42,442	€ 12,442.00		€ 10,000		€ 10,000.00		€ 10,000				€ 42,442	€ 0	€ 42,442.00
TU Delft/Faculty of applied Sciences/Dept of Chemical Engineering	consumables consumables postdoc WP3		€ 39,940			€ 19,940.00		€ 10,000		€ 10,000.00				€ 39,940	€ 0	€ 39,940.00
Erasmus Medisch Centrum/ Dept of Neuroscience	consumables PhD WP 1-2		€ 39,940	€ 12,442.00		€ 10,000		€ 10,000.00		€ 10,000				€ 42,442	€ 0	€ 42,442.00
Erasmus Medisch Centrum/ Dept of Neuroscience	consumables postdoc WP1-3		€ 39,940			€ 19,940.00		€ 10,000		€ 10,000.00				€ 39,940	€ 0	€ 39,940.00
Erasmus Medisch Centrum/ Dept of Neuroscience	consumables postdoc WP2-3		€ 39,940					19,940.00		€ 10,000		€ 10,000.00		€ 39,940	€ 0	€ 39,940.00
TU Delft/Faculty of applied sciences/Dept of imaging physics	consumables matching PhDs WP1		€ 140,000	€ 60,000.00		€ 20,000		€ 20,000.00		€ 20,000		€ 20,000.00		€ 140,000	€ 0	€ 140,000.00
TU Delft/Faculty of applied sciences/Dept of imaging physics	consumables &equipment matching PDs WP1-2		€ 220,000			€ 20,000		€ 140,000.00		€ 40,000		€ 20,000.00		€ 220,000	€ 0	€ 220,000.00
TU Delft/Faculty of applied sciences/Dept of Bionanoscience	consumables matching PhD WP1		€ 20,000	€ 5,000.00		€ 5,000		€ 5,000.00		€ 5,000				€ 20,000	€ 0	€ 20,000.00
TU Delft/Faculty of applied sciences/Dept of Bionanoscience	equipment matchin PhD WP1		€ 100,000	€ 100,000.00										€ 100,000	€ 0	€ 100,000.00
TU Delft/Faculty of applied sciences/Dept of imaging physics	consumable matching PhDs WP1		€ 40,000	€ 5,000.00		€ 10,000		€ 10,000.00		€ 10,000		€ 5,000.00		€ 40,000	€ 0	€ 40,000.00
TU Delft/Faculty of applied sciences/Dept of Bionanoscience	consumable matching PhD WP1		€ 15,000	€ 5,000.00		€ 5,000		€ 5,000.00						€ 15,000	€ 0	€ 15,000.00
TU Delft/Faculty of applied sciences/Dept of Bionanoscience	consumables matching PhDs WP1		€ 30,000			€ 10,000		€ 10,000.00		€ 10,000				€ 30,000	€ 0	€ 30,000.00
TU Delft/Faculty of EEMCS/Dept of microelectronics	consumable matching PhD WP3		€ 60,000			€ 20,000		€ 20,000.00		€ 20,000				€ 60,000	€ 0	€ 60,000.00
TU Delft/Faculty of EEMCS/Dept of microelectronics	consumable matching PhD WP3		€ 50,000			€ 10,000		€ 15,000.00		€ 15,000		€ 10,000.00		€ 50,000	€ 0	€ 50,000.00
TU Delft/Faculty of EEMCS/Dept of microelectronics	consumables matching PhDs&postdoc WP3		€ 200,000	€ 17,500.00		€ 50,000		€ 50,000.00		€ 50,000		€ 32,500.00		€ 200,000	€ 0	€ 200,000.00
Erasmus Medisch Centrum/ Dept of Neuroscience	consumables matching PhD WP1-3		€ 80,000	€ 20,000.00		€ 20,000		€ 20,000.00		€ 10,000		€ 10,000.00		€ 80,000	€ 0	€ 80,000.00
Erasmus Medisch Centrum/ Dept of Neuroscience	consumables matching PhD WP2		€ 10,000	€ 5,000.00		€ 5,000								€ 10,000	€ 0	€ 10,000.00
Erasmus Medisch Centrum/ Dept of Neuroscience	consumables matching Postdoc WP2		€ 10,000	€ 5,000.00		€ 5,000								€ 10,000	€ 0	€ 10,000.00
Erasmus Medisch Centrum/ Dept of Neuroscience	consumables matching postdoc WP2		€ 20,000			€ 5,000		€ 5,000.00		€ 5,000		€ 5,000.00		€ 20,000	€ 0	€ 20,000.00
Erasmus Medisch Centrum/ Dept of Neuroscience	consumables matching PhD WP2		€ 20,000			€ 5,000		€ 5,000.00		€ 5,000		€ 5,000.00		€ 20,000	€ 0	€ 20,000.00
Erasmus Medisch Centrum/ Dept of Neuroscience	consumables matching PhDs WP2		€ 25,000	€ 10,000.00		€ 5,000		€ 5,000.00		€ 5,000				€ 25,000	€ 0	€ 25,000.00
Erasmus Medisch Centrum/ Dept of Neuroscience	consumables matching Postdoc WP3		€ 10,000			€ 5,000		€ 5,000.00						€ 10,000	€ 0	€ 10,000.00

Erasmus Medisch Centrum/ Dept of Neuroscience	consumables matching phd WP3	€ 15,000		€ 5,000	€ 5,000.00	€ 5,000		€ 15,000	€ 0	€ 15,000.00
Erasmus Medisch Centrum/ Dept of Neuroscience	consumables matching phd WP2	€ 20,000	€ 5,000.00	€ 5,000	€ 5,000.00	€ 5,000		€ 20,000	€ 0	€ 20,000.00
Erasmus Medisch Centrum/ Dept of Neuroscience	consumables matching PDs WP2	€ 30,000			€ 10,000.00	€ 10,000	€ 10,000.00	€ 30,000	€ 0	€ 30,000.00
Ondernemingen										
Onderneming 1								€ 0	€ 0	€ 0.00
Onderneming 2								€ 0	€ 0	€ 0.00
Onderneming 3								€ 0	€ 0	€ 0.00
Onderneming 4								€ 0	€ 0	€ 0.00
Overige (private) partners										
Naam 1								€ 0	€ 0	€ 0.00
Naam 2								€ 0	€ 0	€ 0.00
Naam 3								€ 0	€ 0	€ 0.00
Naam 4								€ 0	€ 0	€ 0.00
Totaal		€ 1,357,142						Totaal kosten materialen en hulpmiddelen	€ 1,359,644	

3.	Kosten van gebruik van machines en apparatuur						2022		2023		2024		2025		2026		Totaal		Totaal per partner
Partner die de kosten maakt		Machine / apparaatuur		Prijs per gebruikseenheid	Gebruiks eenheden	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																			
Organisatie 1																	€ 0	€ 0	€ 0.00
Organisatie 2																	€ 0	€ 0	€ 0.00
Organisatie 3																	€ 0	€ 0	€ 0.00
Organisatie 4																	€ 0	€ 0	€ 0.00
Ondernemingen																			
Onderneming 1																	€ 0	€ 0	€ 0.00
Onderneming 2																	€ 0	€ 0	€ 0.00
Onderneming 3																	€ 0	€ 0	€ 0.00
Onderneming 4																	€ 0	€ 0	€ 0.00
Overige (private) partners																			
Naam 1																	€ 0	€ 0	€ 0.00
Naam 2																	€ 0	€ 0	€ 0.00
Naam 3																	€ 0	€ 0	€ 0.00
Naam 4																	€ 0	€ 0	€ 0.00
Totaal						€ 0											Totaal kosten machines en apparatuur		€ 0
4.	Aan derden verschuldigde kosten						2022		2023		2024		2025		2026		Totaal		Totaal per partner
Partner die de kosten maakt		Naam derde, omschrijving kosten			Kosten		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																			
TU Delft/Faculty of applied sciences/Dept of imaging physics		kosten educatie, outreach, compliance			€ 23,334	€	4,670.00		€ 4,666		€ 4,666.00		€ 4,666		€ 4,666.00		€ 23,334	€ 0	€ 23,334.00
Erasmus Medisch Centrum/ Dept of Neuroscience		kosten educatie, outreach, compliance			€ 23,334	€	4,670.00		€ 4,666		€ 4,666.00		€ 4,666		€ 4,666.00		€ 23,334	€ 0	€ 23,334.00
Erasmus Universiteit Rotterdam/Dept of Philosophy		kosten educatie, outreach, compliance			€ 23,334	€	4,670.00		€ 4,666		€ 4,666.00		€ 4,666		€ 4,666.00		€ 23,334	€ 0	€ 23,334.00
Organisatie 4																	€ 0	€ 0	€ 0.00
Ondernemingen																			
Onderneming 1																	€ 0	€ 0	€ 0.00
Onderneming 2																	€ 0	€ 0	€ 0.00
Onderneming 3																	€ 0	€ 0	€ 0.00
Onderneming 4																	€ 0	€ 0	€ 0.00
Overige (private) partners																			
Naam 1																	€ 0	€ 0	€ 0.00
Naam 2																	€ 0	€ 0	€ 0.00
Naam 3																	€ 0	€ 0	€ 0.00
Naam 4																	€ 0	€ 0	€ 0.00
Totaal						€ 70,002											Totaal kosten derden		€ 70,002
5.	Publicatie, reis- en verblijfkosten						2022		2023		2024		2025		2026		Totaal		Totaal per partner
Partner die de kosten maakt		Naam derde, omschrijving kosten			Kosten		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																			
TU Delft/Faculty of applied sciences/Dept of imaging physics		publicatie, reis en verblijfkosten			€ 10,000	€	3,000.00		€ 3,000		€ 4,000.00						€ 10,000		€ 10,000.00
TU Delft/Faculty of applied sciences/Dept of Bionanoscience		publicatie, reis en verblijfkosten			€ 10,000	€	2,000.00		€ 2,000		€ 2,000.00		€ 4,000				€ 10,000		€ 10,000.00
TU Delft/Faculty of applied Sciences/Dept of Chemical Engineering		publicatie, reis en verblijfkosten			€ 10,000				€ 2,000		€ 2,000.00		€ 2,000		€ 4,000.00		€ 10,000		€ 10,000.00
Erasmus Medisch Centrum/ Dept of Neuroscience		publicatie, reis en verblijfkosten			€ 10,000	€	3,000.00		€ 3,000		€ 4,000.00						€ 10,000		€ 10,000.00
Erasmus Medisch Centrum/ Dept of Neuroscience		publicatie, reis en verblijfkosten			€ 10,000				€ 3,000		€ 3,000.00		€ 4,000				€ 10,000		€ 10,000.00
Erasmus Medisch Centrum/ Dept of Neuroscience		publicatie, reis en verblijfkosten			€ 10,000				€ 3,000		€ 3,000.00		€ 3,000		€ 4,000.00		€ 10,000		€ 10,000.00
Erasmus Universiteit Rotterdam/Dept of Philosophy		publicatie, reis en verblijfkosten			€ 22,418	€	5,000.00		€ 5,000		€ 5,000.00		€ 7,418				€ 22,418		€ 22,418.00
Ondernemingen																			
Onderneming 1																	€ 0	€ 0	€ 0.00
Onderneming 2																	€ 0	€ 0	€ 0.00
Onderneming 3																	€ 0	€ 0	€ 0.00
Onderneming 4																	€ 0	€ 0	€ 0.00
Overige (private) partners																			
Naam 1																	€ 0	€ 0	€ 0.00
Naam 2																	€ 0	€ 0	€ 0.00
Naam 3																	€ 0	€ 0	€ 0.00
Naam 4																	€ 0	€ 0	€ 0.00
Totaal						€ 82,418											Totaal kosten derden		€ 82,418
6.	Totale kosten						€ 11,071,084										€ 11,073,585		

FINANCIERING - Bijdragen per partner per jaar

In Euro		2022		2023		2024		2025		2026		Totaal		Totaal per partner
		In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	
Onderzoeksorganisaties														
1	TU Delft/Faculty of AS/Dept of Imaging physics	€ 272,084		€ 429,448		€ 492,088		€ 277,352		€ 177,360		€ 1,648,332	€ 0	€ 1,648,332
2	TU Delft/Faculty of AS/Dept of Bionanoscience	€ 175,234		€ 150,468		€ 150,468		€ 125,390		€ 25,078		€ 626,638	€ 0	€ 626,638
3	TU Delft/Faculty of 3ME/Delft center for systems and control	€ 106,071		€ 223,405		€ 262,098		€ 319,468		€ 299,472		€ 1,210,514	€ 0	€ 1,210,514
4	TU Delft/Faculty of EEMCS/Dept of microelectronics	€ 203,914		€ 499,956		€ 404,570		€ 390,233		€ 200,262		€ 1,698,935	€ 0	€ 1,698,935
5	TU Delft/Faculty of AS/Dept of Chemical Engineering	€ 133,124		€ 99,669		€ 57,362		€ 57,368		€ 57,368		€ 404,891	€ 0	€ 404,891
6	Erasmus Medisch Centrum/ Dept of Neuroscience	€ 509,261		€ 879,758		€ 777,244		€ 629,108		€ 459,440		€ 3,254,811	€ 0	€ 3,254,811
7	Erasmus Universiteit Rotterdam/Dept of Clinical Psycholoog	€ 57,360		€ 57,368		€ 57,368		€ 57,368				€ 229,464	€ 0	€ 229,464
8	Convergence Health and Technology	€ 400,000		€ 400,000		€ 400,000		€ 400,000		€ 400,000		€ 2,000,000	€ 0	€ 2,000,000
	Totale bijdrage onderzoeksorganisaties	€ 1,857,048	€ 0	€ 2,740,072	€ 0	€ 2,601,198	€ 0	€ 2,256,287	€ 0	€ 1,618,980	€ 0	€ 11,073,585	€ 0	€ 11,073,585
Ondernemingen														
4	Onderneming 1											€ 0	€ 0	€ 0
5	Onderneming 2											€ 0	€ 0	€ 0
6	Onderneming 3											€ 0	€ 0	€ 0
...	Onderneming 4											€ 0	€ 0	€ 0
	Totale bijdrage ondernemingen	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0
Overige (private) partners														
7	Naam 1											€ 0	€ 0	€ 0
8	Naam 2											€ 0	€ 0	€ 0
9	Naam 3											€ 0	€ 0	€ 0
...	Naam 4											€ 0	€ 0	€ 0
	Totale overige bijdrage	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0
		In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind			Totaal
Totaal budget per jaar		€ 1,857,048	€ 0	€ 2,740,072	€ 0	€ 2,601,198	€ 0	€ 2,256,287	€ 0	€ 1,618,980	€ 0			€ 11,073,585
														Funding gap
Totaal budget		€ 1,857,048		€ 2,740,072		€ 2,601,198		€ 2,256,287		€ 1,618,980		€ 11,073,585		

Annex II Letters of intent and letters of commitment:

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

Dr. Daan Brinks
TU Delft

Phone 0031 (0)10 7030051
E-mail t.barakat@erasmusmc.nl
Our reference 2022-23
Date 09/03/2022

Concerning LETTER OF INTENT for the INTEGRATIVE NEUROMEDICINE PROJECT

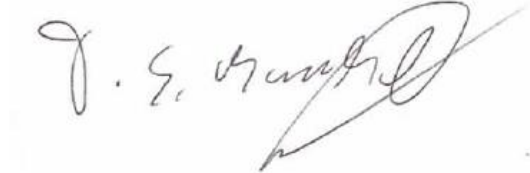
Dear Daan,

I, Stefan Barakat, in my capacity of assistant professor at the Erasmus MC, hereby confirm that I intend to contribute to the Integrative Neuromedicine project, as applied for in the proposal to which this letter is attached.

I intend to contribute through my granted EpilepsieNL grant 2021 and CURE Epilepsy Award 2022, which are well aligned with the Flagship program as both focus on therapy development for epilepsy. From these grant, I intend to contribute € 100,000 in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Moreover, I have submitted a ZonMw Vidi and intend to apply for an ERC-STG in 2022 which will similarly align with the proposed flagship program goals, and which, if granted, will result in additional matching for this initiative, which I am more than happy to join and support.

Sincerely,



Dr. T.S. Barakat PhD, MD, MSc
Assistant professor
Head of the non-coding genome research group
Clinical Genetics

Postal address
P.O. box 2040
3000 CA Rotterdam, NL

Visiting address
Dr. Molewaterplein 40
3015 GD Rotterdam, NL

Contact & route
www.erasmusmc.nl



G.M. van Woerden, PhD
Associate Professor Mol. Neuroanatomy
Dept of Neuroscience
Dept of Clinical Genetics
PO box 2040
3000 CA Rotterdam
Direct dial: +31-10 703 2695
Email: g.vanwoerden@erasmusmc.nl

Date: March 10, 2022

LETTER OF INTENT
for the
INTEGRATIVE NEUROMEDICINE PROJECT

I, Geeske van Woerden, in my capacity of Associate Professor at the Erasmus MC hereby confirm that I intend to contribute to the Integrative Neuromedicine project, as applied for in the proposal to which this letter is attached.

I intend to contribute through my VIDI grant "*unraveling the function of CAMK2 in brain development and its role in neurodevelopmental disorders*" (0.16.Vidi.188.014) for which I developed different techniques (such as the MEAs) that are very well aligned with the project proposed here. From this grant I intend to contribute € 240.000 in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Yours sincerely,

Name: Geeske M. van Woerden
Position: Associate Professor
Date: 10-03-2022

LETTER OF INTENT
for the
INTEGRATIVE NEUROMEDICINE PROJECT

In this letter of intent I state my full commitment to the Integrative Neuromedicine project, as applied for in the proposal to which this letter is attached.

I intend to contribute through my H2020 CEol grant “ArborIO”, which is well aligned with the Flagship program as it focuses on enabling technologies to model human-scale biophysically plausible neural networks, and particularly, the cerebellar network.

From this grant, I intend to contribute € 250k in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Moreover, I intend to apply in 2022 for a Human Frontiers researcher grant, focusing on memristor technology applied for the modeling of biological neural networks, with the aim of contributing to the flagship another €500k in cash in the period 2023 –2027.

Postal address

P.O. box 2040
3000 CA Rotterdam, NL

Visiting address

Dr. Molewaterplein 40
3015 GD Rotterdam, NL

Contact & route

www.erasmusmc.nl



Mario Negrello, PhD

Assistant professor

Erasmus Medical Center / Technical University Delft

Date 11/03/2022
Name Tiago Costa
Faculty Electrical Engineering, Mathematics and Computer Science
Department Microelectronics
Address Mekelweg 4, 2628 CD Delft
Telephone +31 152789504
Cell +31 627826131
E-mail t.m.l.dacosta@tudelft.nl
Website <https://www.tmlcosta.com/>



LETTER OF INTENT
for the
Integrative Neuromedicine PROJECT

I, Tiago Costa, in my capacity of Assistant Professor at Delft University of Technology hereby confirm that I intend to contribute to the Integrative Neuromedicine project, as applied for in the proposal to which this letter is attached.

I intend to contribute through my granted HFSP young investigator grant – unraveling the fundamental mechanisms of neuromodulation by focused ultrasound; through the EWI-Microelectronics synergy grant ICARUS – a brain-on-chip platform for focused ultrasonic stimulation; and through the TUD start-up package – epidural ultrasonic brain stimulation. All of the above align very well with the Flagship program by involving the research of novel focused ultrasound technology for both fundamental studies and therapeutic devices.

From these grants, I intend to contribute € 533270 in cash towards the projects costs in accordance with the budget in the project proposal and budget form.

Moreover, I have applied for the EIC Pathfinder Challenges: Tools to measure & stimulate activity in brain tissue (results in 2022) with the UPSIDE project, focusing on epidural ultrasonic deep brain stimulation for the treatment of depression; and for the NWO Zwaartekracht (results in 2022) with the DBI² (The Dutch Brain Interface Initiative), for which my contribution will focus on ultra-high spatial-resolution focused-ultrasound devices for neuroscience. From these projects, I intend to contribute to the flagship another €886201 in cash in the period 2022-2026.

Yours sincerely,

Name: Tiago Costa
Position: Assistant Professor
Date: 11-3-2022

A handwritten signature in blue ink, appearing to read "Costa Tiago", with a horizontal line drawn underneath it.

Concerning: LETTER OF INTENT for the INTEGRATIVE NEUROMEDICINE PROJECT

Postal address
P.O. box 2040
3000 CA Rotterdam, NL

Street address
Wytemaweg 80
3015 CN Rotterdam, NL

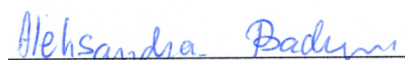
I, Aleksandra Badura, in my capacity of Associate Professor at the Erasmus MC hereby confirm that I intend to contribute to the Integrative Neuromedicine project, as applied for in the proposal to which this letter is attached.

I intend to contribute through my VIDI grant “Cerebello-cerebral networks underlying shared autistic traits” (Vidi.917.18.380,2018), for which I have established novel methods to measure autistic traits in mouse models (such as the functional ultrasound) that are very well aligned with the project proposed here. I will also contribute through my Horizon 2020-ITN PIPgen grant, which supports a PhD student working on the contribution of the mTOR pathway to the development of ASD. Under this grant we collaborate with a pharmaceutical company Pharming on investigating the possible treatment options, which is part of work package 3 of the Integrative Neuromedicine Proposal.

From these grants I intend to contribute €311,840 in cash towards the project costs in accordance with the budget in the project proposal and the budget form.

Moreover, I intend to apply for an HFSP research grant in 2023, focusing on investigation of mechanisms of neurodevelopmental deficits in primary immune deficiency, and an ENW-M in 2022, focusing on the inhibition-excitation balance in ASD, with the aim of contributing to the flagship another €348,616 in cash in the period 2023 –2027.

Yours sincerely,



Dr. Aleksandra Badura
Associate Professor
Department of Neuroscience

LETTER OF INTENT
for the
INTEGRATIVE NEUROMEDICINE PROJECT

I, Daan Brinks, in my capacity of Assistant Professor at Delft University of Technology hereby confirm that I intend to contribute to the Integrative Neuromedicine project, as applied for in the proposal to which this letter is attached.

I intend to contribute through my granted ERC starting grant, 850418, Multivision - Multiphoton Voltage Imaging, which is well aligned with the Flagship program as it focuses on development of optical assays to evaluated neural activity; and through my granted CSC project Nanoscopic Optogenetics, which is well aligned with the Flagship program as it focuses on the development of assays for measuring synaptic activity in disease models.

From these grants, I intend to contribute € 598944 in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Moreover, I intend to apply for an HFSP young researcher grant, focusing on in vivo investigation of mechanisms of formation of aberrant cell types in neurodegenerative disease, in 2022 and an ENW-M, focusing on in vivo voltage imaging, in 2022, with the aim of contributing to the flagship another € 700000 in cash in the period 2023 –2027.

Yours sincerely,

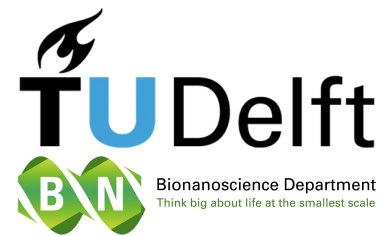


Name: Daan Brinks

Position: Assistant Professor

Date: 6-3-2022

Dr. Dimphna H. Meijer
Assistant Professor
Bionanoscience Department
Delft University of Technology
Van der Maasweg 9, 2629 HZ Delft
The Netherlands
E-mail: D.H.M.Meijer@tudelft.nl



Delft, 09 March 2022

LETTER OF INTENT
for the
Integrative Neuromedicine PROJECT

I, Dimphna Meijer, in my capacity of assistant professor and group leader at Delft University of Technology hereby confirm that I intend to contribute to the Integrative Neuromedicine project, as applied for in the proposal to which this letter is attached.

I intend to contribute through my granted TUD start-up package, which is well aligned with the Flagship program as it focuses on understanding molecular mechanisms of the developing brain. From this grant, I intend to contribute €[75k in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Moreover, I intend to apply for the ERC stg grant and NWO VIDI grant , both focusing on developing novel technological tools to understand healthy brain function in 2022/2023, with the aim of contributing to the flagship another € 200k in cash in the period 2022–2026.

Yours sincerely,

A handwritten signature in blue ink, appearing to read 'D Meijer'.

Name: Dimphna Meijer
Position: Assistant professor and group leader
Date: 09 March 2022

LETTER OF INTENT
for the
INTEGRATIVE NEUROMEDICINE PROJECT

I, Alina Rwei, in my capacity of Assistant Professor at Delft University of Technology hereby confirm that I intend to contribute to the Integrative Neuromedicine project, as applied for in the proposal to which this letter is attached.

I intend to contribute through my granted CSC project ultrasound-triggerable drug delivery which is well aligned with the Flagship program as it focuses on the development of externally-triggerable drug delivery systems. I also intend to contribute through my granted start-up package on the project biomimetic, targeted drug delivery which is well aligned with the Flagship program as it focuses on the development of brain-targeting drug delivery systems.

From these grants, I intend to contribute €232783 in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Moreover, I intend to apply for ENW-M1, focusing on the development of externally-triggerable drug delivery systems targeting the mTOR pathway in 2024, with the aim of contributing to the flagship another € 172104 in cash in the period 2024-2027.

Yours sincerely,

A handwritten signature in black ink that reads 'Alina Rwei' in a cursive script.

Name: Alina Rwei

Position: Assistant Professor

Date: 11-03-2022

Delft, 11.03.2022

LETTER OF INTENT
for the
INTEGRATIVE NEUROMEDICINE PROJECT

I, Kristin Großmayer, in my capacity of Assistant Professor at TU Delft hereby confirm that I intend to contribute to the Integrative Neuromedicine project, as applied for in the proposal to which this letter is attached.

I intend to contribute through my TU Delft Startup Funding, which is well aligned with the Flagship program as it focuses on development and application of super-resolution imaging for cell biology.

From this grant, I intend to contribute € 200000 in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Moreover, I applied for an NOW-Vidi grant “Neurodegenerative disease under the magnifying glass” in 2021, and I am planning an ERC StG focusing on the development of fast, live-cell compatible functional 3D super-resolution imaging in 2022 with the aim of contributing to the flagship another € 200000 in cash in the period 2022 – 2026.

Yours sincerely,



Ass. Prof. dr. Kristin Großmayer

TU Delft / Department of Bionanoscience
Kavli Institute of Nanoscience
Faculty of Applied Sciences
Delft University of Technology

LETTER OF INTENT
for the
INTEGRATIVE NEUROMEDICINE PROJECT

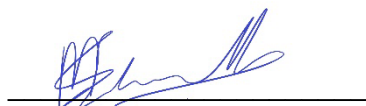
I, Martijn Schonewille, in my capacity of Associate Professor at Erasmus MC hereby confirm that I intend to contribute to the Integrative Neuromedicine project, as applied for in the proposal to which this letter is attached.

I intend to contribute through my granted DfG Fellowship for dr. Friederike Stephani (Postdoc Fellowship, extended to 42 months; project title: "*Characterization of dissociated Purkinje neurons*") and a NRF-Nuffic Fellowship for Heiling Lu (PhD Fellowship, 48 months; project title: "*Identification and Evaluation of the Purkinje cell-related Factors Responsible for The Neurodegeneration in Spino-Cerebellar Ataxias*"), which are well aligned with the Flagship program as it focuses on culturing primary neurons, particularly cerebellar Purkinje cells, and studying the role of mTOR-related proteins in cerebellar development.

From this grant, I intend to contribute, via consumables + salaries (based on equivalent costs) € 286248 in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Moreover, I intend to apply for an ENW-M grant, focusing on cerebellar development of sensorimotor integration in 2022 and VICI, focusing on the role of cerebellar differentiation in health and disease in 2023, with the aim of contributing to the flagship another € 1850000 in cash in the period 2023 – 2027.

Yours sincerely,



Name: Martijn Schonewille

Position: Associate Professor

Date: 8-3-2022

Delft University of Technology
Faculty of Mechanical, Maritime, and Materials
Engineering
Department of Delft Center for Systems & Control

Mekelweg 2, 3mE
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11 March 2022

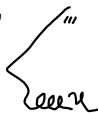
LETTER OF INTENT
for the
INTEGRATIVE NEUROMEDICINE PROJECT

I, Matin Jafarian, in my capacity of tenure-track Assistant Professor at Delft University of Technology hereby confirm that I intend to contribute to the Integrative Neuromedicine project, as applied for in the proposal to which this letter is attached.

I intend to contribute through my granted TUD Start Ups, which is well aligned with the Flagship program as it focuses on cortical level modelling, analysis and regulation of memory. From this grant, I intend to contribute € 70 K in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Moreover, I intend to apply for NWO ViDi and ENM-M in 2022-23, focusing on micro level modelling in human memory system, with the aim of contributing to the flagship another € 300K in cash in the period 2023-2027.

Yours sincerely,



Name: Matin Jafarian

Position: Assistant Professor

Date: 11-03-2022

Annex III Letter of support from department head of the main lead:

On headed paper of the department or institution.

Prof.dr. S. Stallinga
Department of Imaging Physics
Faculty of Applied Sciences

Lorentzweg 1
2628 CJ Delft

T: +31-15-2783538
M: +31-06-20982787
E: s.stallinga@tudelft.nl

To whom it may concern,

Herewith I want to give my full, wholehearted support for the health-tech convergence proposal for a Flagship on Integrative Neuromedicine, led by dr.ir. D. Brinks, assistant professor in my department of Imaging Physics at TU Delft. I am duly impressed by the depth and forward looking vision of this group of predominantly young scientists at our universities. The match with the goals of health-tech convergence, transforming the way we work on healthcare in the broadest sense, could hardly be better. My hope is that the initiative of this young consortium will receive the support from our universities to accelerate and sustain its growth to international leadership in the field of neuromedicine.

With kind regards,

Prof. dr. S. Stallinga
Digitally signed by
Prof. dr. S. Stallinga
Date: 2022.03.09
20:38:05 +01'00'

prof. dr. Sjoerd Stallinga
Head of Department of Imaging Physics, faculty of Applied Sciences, TU Delft

Annex IV References

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